A

Major Project Report on

POWER QUALITY DISTURBANCE DETECTION AND CLASSIFICATION USING MACHINE LEARNING

Submitted in partial fulfilment of the requirements for the curriculum of

Bachelor of Engineering in Electrical Engineering (Sem VII–VIII)

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CERTIFICATE

This is to certify that the Major Project entitled **"Power Quality Disturbance Detection and Classification Using Machine Learning and Deep Representation Learning"** is a bonafide record of work carried out by **Allen Salmo Nadar (1023169), Ryan Norman (1023170), Tushar Rajendra Patil (4023126), and Sejal Sampat Pawar (4023128)** under our supervision, submitted in partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering in Electrical Engineering (Sem VII–VIII) from Fr. C. Rodrigues Institute of Technology, Vashi (affiliated to the University of Mumbai) during the academic year 2026–2027.

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Declaration

We declare that this written project report entitled **"Power Quality Disturbance Detection and Classification Using Machine Learning and Deep Representation Learning"** represents our own original work completed during our final-year engineering curriculum. Where the work, data, equations, or publications of others have been utilized, they have been properly cited and referenced in accordance with academic integrity guidelines.

We confirm that all experimental evaluations, dataset counts, and empirical findings reported herein are derived honestly from the project's simulation models and reproducible codebases, without data falsification or fabrication. We further acknowledge that any breach of academic integrity is subject to disciplinary action by the Institute.

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Abstract

Power quality (PQ) monitoring has emerged as a paramount technical requirement in modern power grids characterized by widespread non-linear loads, high-frequency power electronic switching converters, and distributed energy resource integrations. Voltage and current waveform disturbances degrade equipment efficiency, induce thermal stress, trip sensitive automated protections, and reduce overall network reliability. Developing automated, high-accuracy, and computationally resilient algorithms for power quality disturbance classification is therefore a crucial engineering priority.

This project establishes a comprehensive, end-to-end framework for three-phase power quality disturbance detection and classification grounded in the Western System Coordinating Council (WSCC) IEEE 9-bus transmission benchmark (60 Hz, 230 kV nominal base). The entire processing pipeline is developed and validated across nine systematic stages: (1) electrical network modeling in MATLAB/Simulink; (2) controlled disturbance injection across eight canonical IEEE Std 1159 classes (Normal, Sag, Swell, Interruption, Harmonics, Flicker, Notch, and Oscillatory Transient); (3) multi-channel three-phase voltage and current signal acquisition at the Bus 5 load center; (4) dual-rate digital signal processing (20 kHz high-resolution and 5 kHz anti-aliased decimated streams) and 32-dimensional canonical feature extraction; (5) rigorous dataset generation and leakage-free partitioning; (6) classical machine learning benchmarking (MLP, SVM, Random Forest, Extra Trees, HistGradientBoosting, Logistic Regression); (7) raw waveform deep learning evaluation utilizing Compact 1D Convolutional Neural Networks and Temporal Convolutional Networks (TCN) with dilated causal receptive fields; (8) multi-modal hybrid representation fusion via **HybridPQNet**; and (9) an interactive research dashboard for recorded-data analytics and explainability.

Empirical evaluations on a strictly frozen, zero-leakage dataset (comprising 4,608 frames across 144 independent continuous trajectories) confirm that while classical feature-based MLP models achieve 92.60% validation Macro F1 and raw-waveform TCN architectures achieve 93.70% validation Macro F1, the fused **HybridPQNet** model achieves **100.00% validation Macro F1** and maintains **100.00% Macro F1** on a quarantined one-time challenge partition spanning four physically held-out grid stress regimes (Renewable Inverter Penetration, Severe Reactive Imbalance, Asymmetric Substation

Loading, Stressed Transmission Import) and harsh 40 dB sensor noise. Group ablation proves that hand-crafted physical features and convolutional waveform representations supply complementary mathematical properties: raw waveforms resolve localized sub-cycle commutation slopes and high-frequency phase coupling, while engineered features supply invariant global energy conservation quantities. All findings reflect simulated electromagnetic benchmark environments, with physical grid deployment and embedded hardware execution demarcated as future scope.

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Chapter 1

Introduction

1.1 Background and Context

Electrical energy is the foundational backbone of industrial automation, critical healthcare facilities, computational data centers, and modern residential infrastructure. In an ideal alternating-current (AC) power system, electrical energy is delivered at a strictly regulated fundamental frequency (60.0 Hz or 50.0 Hz) and a balanced, pure sinusoidal voltage waveform across all three phases. However, modern electrical distribution and transmission networks have departed significantly from linear, passive characteristics.

The proliferation of non-linear industrial loads, including adjustable speed motor drives (ASDs), electric arc furnaces, high-power rectifiers, switched-mode power supplies (SMPS), and distributed inverter-based renewable energy resources (such as photovoltaic arrays and wind generators), injects substantial harmonic currents and switching transients into the grid. Concurrently, routine operational events—such as large induction motor starts, shunt capacitor bank energization, transformer inrush, and lightning strikes or short-circuit faults on transmission lines—distort system voltage and current waveforms. These deviations are collectively categorized as Power Quality Disturbances (PQDs).

Recognizing and mitigating PQDs is vital because voltage sags cause sensitive programmable logic controllers (PLCs) and contactors to drop out, voltage swells degrade insulation and damage surge arresters, sustained harmonics increase copper losses and induce transformer overheating, and commutation notches cause thyristor misfiring. Traditional PQ monitoring relies on manual visual inspection of oscillographic records by utility engineers, a procedure that is prohibitively slow, labor-intensive, and inadequate for modern smart grid automated control. Automated, machine-learning-based PQ disturbance detection and classification provides an intelligent, continuous surveillance paradigm capable of identifying event signatures within milliseconds.

1.2 Motivation

Although automated power quality classification has been researched for over two decades, substantial gaps persist between academic proofs-of-concept and robust engineering implementations:

1. **Single-Phase vs. Synchronized Three-Phase Dynamics:** A significant portion of existing literature evaluates machine learning models on synthetic, single-phase mathematical equations ($v(t) = A\sin(\omega t)$). Real electrical transmission grids operate in balanced three-phase configurations where asymmetrical faults (single-line-to-ground, line-to-line) alter inter-phase voltage unbalance ($V_{\text{unbalance}}$) and neutral currents. Evaluating models without synchronized three-phase multi-channel acquisition masks these physical realities.

2. **High-Frequency Sub-Cycle Transients vs. Sampling Constraints:** Disturbances such as commutation notches (durations of 200 to 500 μs) and capacitor switching transients (frequencies of 1.2 to 2.2 kHz) exhibit fast transient dynamics. Decimating signals to low sampling rates (1 to 5 kHz) without proper anti-aliasing blurs sharp dv/dt boundaries, confounding transient spikes with background measurement noise.

3. **The Feature vs. End-to-End Representation Dilemma:** Classical machine learning relies on carefully hand-crafted feature vectors (RMS, crest factor, Goertzel harmonic bins), which discard temporal phase morphology. Conversely, modern end-to-end deep learning networks (1D CNNs, LSTMs) operate on raw waveforms but struggle to compute global mathematical integrals (such as true RMS and THD) without millions of parameters. Determining whether engineered features and deep waveform representations are mutually redundant or mathematically complementary is an urgent research question.

4. **Operating-Point Sensitivity & Leakage-Free Benchmarks:** Many reported high-accuracy classifiers fail when tested under varying power flow regimes, heavy inductive loading, or high sensor noise because their datasets leak operating conditions across training and test splits.

1.3 Aim and Objectives

The primary aim of this final-year project is to design, implement, and rigorously benchmark a multi-modal power quality disturbance detection and classification system utiliz-

ing the IEEE 9-bus WSCC transmission benchmark model in MATLAB/Simulink, evaluating classical machine learning, raw waveform deep learning, and hybrid multi-modal fusion architectures under zero-leakage conditions.

The specific research and engineering objectives are:

- **Simulink Benchmark Modeling:** Implement the WSCC 3-machine 9-bus system (60 Hz, 230 kV) in MATLAB/Simulink with dedicated disturbance injection blocks for eight IEEE Std 1159 classes (Normal, Sag, Swell, Interruption, Harmonics, Flicker, Notch, Transient).

- **Multi-Rate Acquisition & Noise Modeling:** Capture synchronized three-phase voltage (V_{abc}) and current (I_{abc}) signals at the Bus 5 load center at 20 kHz and 5 kHz (via an 8th-order Chebyshev Type I anti-alias filter), incorporating calibrated fixed-reference sensor noise (52 dB, 46 dB, and 40 dB SNR) and 12-bit/14-bit bipolar ADC quantization.

- **Canonical Feature Engineering:** Extract a standardized 32-dimensional feature contract spanning time-domain amplitude, signal-to-noise ratio via orthogonal fundamental projection, Goertzel harmonic magnitudes (H1–H11), odd harmonic ratios, and spectral moments.

- **Leakage-Free Dataset Curation:** Generate a frozen, group-aware dataset partition ensuring that no continuous simulation trajectory or operating condition is shared across training, validation, and challenge subsets.

- **Classical ML Benchmarking & Feature Ablation:** Train and benchmark six classical classifiers (MLP, SVM, Random Forest, Extra Trees, HistGradientBoosting, Logistic Regression) and perform systematic functional group ablation to establish feature criticality.

- **Raw Waveform Deep Learning:** Develop and evaluate Compact 1D CNNs and Temporal Convolutional Networks (TCN) directly on multi-channel raw waveforms, quantifying the impact of three-phase synchrony and sampling rate.

- **Hybrid Representation Learning (HybridPQNet):** Architect a dual-branch neural network that fuses convolutional waveform embeddings with normalized physical feature vectors, evaluating out-of-distribution robustness across held-out grid stress conditions.

- **Interactive Dashboard Implementation:** Deploy a user-facing analytics dashboard in Streamlit for recorded-data exploration and model explainability.

1.4 Problem Definition

Given a continuous three-phase electrical voltage signal $\mathbf{v}(t) = [v_a(t), v_b(t), v_c(t)]^T$ monitored at a nominal transmission bus over an observation window of $T = 200$ ms (corresponding to 12 cycles at 60 Hz), the objective is to accurately map the signal to one of eight discrete mutually exclusive operational states:

$$y \in \mathcal{C} = \{\text{Flicker (0), Harmonics (1), Interruption (2), Normal (3), Notch (4), Sag (5), Swell (6), Trans} \tag{1.1}$$

The classification algorithm must remain invariant to: (a) variations in grid active and reactive load operating points; (b) phase fault asymmetries; (c) localized voltage offsets ($V_{\text{bus}} \neq 1.0$ pu); and (d) additive measurement sensor noise down to 40 dB SNR.

1.5 Scope and Boundaries of the Work

To maintain academic integrity, the technical scope is defined as follows:

- **Within Scope:** Rigorous electromagnetic transient simulation on the IEEE 9-bus WSCC benchmark using the stiff `ode23tb` solver in MATLAB/Simulink; physical fault modeling at Bus 4 and Bus 5; dual-rate digital signal processing; feature engineering and mathematical Goertzel filtering; classical and deep neural network training; offline and recorded-data dashboard visualization; and automated regression testing.
- **Future Scope (Phase 6):** Embedded edge deployment on physical microcontrollers (e.g., ESP32, STM32, ARM Cortex-M), closed-loop Hardware-in-the-Loop (HIL) execution, physical electrical sensor interfacing, and live transmission grid deployment. The present work does not claim physical utility grid deployment.

Chapter 2

Literature Review

2.1 Overview of Power Quality Standards

Power quality disturbances are governed internationally by two primary IEEE standards:

- **IEEE Std 1159-2019 (Recommended Practice for Monitoring Electric Power Quality):** Categorizes electromagnetic phenomena into transients (impulsive and oscillatory), short-duration variations (sag, swell, interruption), long-duration variations, voltage unbalance, waveform distortions (harmonics, interharmonics, notch, noise), voltage fluctuations (flicker), and power frequency variations [1]. Table 2.1 summarizes the threshold definitions relevant to this study.
- **IEEE Std 519-2022 (Standard for Harmonic Control in Electric Power Systems):** Establishes limits on voltage and current harmonic distortion at the Point of Common Coupling (PCC), specifying that maximum individual harmonic voltage distortion must remain below 3.0% and Total Harmonic Distortion (THD) below 5.0% for systems under 69 kV [2].

Table 2.1: Power-Quality Disturbance Classification Criteria Based on IEEE Standards

Disturbance Class	**Typical Duration**	**Voltage Magnitude (pu)**	**Spectral / Waveform Signature**
Normal Operation	Continuous	$0.90 \leq V_{\text{rms}} \leq 1.10$	Nominal 60 Hz, THD $<$ 5%
Voltage Sag	0.5 cycles to 1 min	$0.10 \leq V_{\text{rms}} < 0.90$	Amplitude reduction
Voltage Swell	0.5 cycles to 1 min	$1.10 < V_{\text{rms}} \leq 1.80$	Amplitude elevation
Voltage Interruption	0.5 cycles to 1 min	$V_{\text{rms}} < 0.10$	Supply collapse
Harmonics	Steady-state	$0.90 \leq V_{\text{rms}} \leq 1.10$	Integer multiples of f_0, THD $> 5\%$
Voltage Flicker	Intermittent	$0.90 \leq V_{\text{rms}} \leq 1.10$	Sub-band AM (3–20 Hz)
Commutation Notch	Steady-state / periodic	$0.90 \leq V_{\text{rms}} \leq 1.10$	Sub-cycle dip (200–500 μs)
Oscillatory Transient	Sub-cycle (< 50 ms)	Peak up to 2.0 pu	Damped ringing (1–5 kHz)

2.2 Review of Contemporary Research Studies

Extensive research has examined data-driven and signal-processing techniques for PQ disturbance recognition:

2.2.1 Artificial Neural Networks with Wavelet Decomposition

Buasi and Srirattanawichaikul (2023) investigated the recognition of power quality disturbances in distribution networks using an Artificial Neural Network (ANN) coupled with Discrete Wavelet Transform (DWT) feature extraction [7]. The authors evaluated multi-resolution decomposition across Daubechies mother wavelets (db4) to capture high-frequency energy components under sag, swell, harmonics, flicker, and transients. While the initial synthetic evaluation reported an accuracy of 99.4%, testing the trained model on an IEEE 9-bus benchmark system caused performance to degrade to 95.6%. This drop highlighted that network topology interactions and multi-bus loading introduce spectral smearing that simple wavelet energy thresholds fail to isolate.

2.2.2 Morphology Filters and Decision Tree Classifiers

Sekar, Sendilkumar, and Karthick (2020) proposed a machine-learning framework utilizing Mathematical Morphology (MM) filters combined with a Decision Tree (DT) classifier [8]. The MM filter applied structural elements to decompose voltage signals into multiscale opening and closing profiles, successfully extracting gradient inflection points for ten disturbance classes with an accuracy of 99.95%. However, their evaluation was

performed purely on single-phase synthetic mathematical waveforms generated without electrical network impedances or sensor measurement noise, leaving open the question of how morphology filters perform under realistic transmission line fault transients.

2.2.3 Fourier Synchro-Squeezing Transform and Ensemble Methods

Shivesh et al. (2024) explored time-frequency representation learning by applying the Fourier Synchro Squeezing Transform (FSST) to preprocess PQ disturbance records prior to classification with classical ensemble algorithms [9]. Evaluating four algorithms—Support Vector Machine (SVM), Random Forest (RF), Gradient Boosting Machine (GBM), and Logistic Regression (LR)—the authors reported accuracies of 91.0%, 96.0%, 99.0%, and 90.0%, respectively. The study underscored that gradient-boosted decision trees handle non-linear time-frequency projections exceptionally well, but FSST computation requires substantial matrix operations that incur latency penalties in real-time edge hardware.

2.2.4 Deep Learning and End-to-End Convolutional Models

Recent literature has increasingly focused on 1D and 2D Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) operating directly on sampled waveforms [10,11]. These models bypass manual feature extraction by learning hierarchical temporal filters. However, standard CNNs have finite receptive fields that struggle to maintain long-range context across multiple fundamental cycles, motivating the investigation of Temporal Convolutional Networks (TCN) with dilated causal convolutions.

2.3 Summary of Literature Review and Identified Research Gaps

Table 2.2 synthesizes the comparative analysis of existing literature and situates the contributions of this project.

Table 2.2: Synthesis of Published Literature on Power-Quality Disturbance Detection and Classification

Study	**Signal Representation**	**Classifier**	**Reported Accuracy**	**Identified Limitations**
Buasi et al. (2023) [7]	DWT Decomposition	Artificial Neural Network	95.6% on IEEE 9-bus	Sensitive to grid loading; single-phase evaluation
Sekar et al. (2020) [8]	Mathematical Morphology	Decision Tree	99.95% (Synthetic)	No network physics; noise-free assumption
Shivesh et al. (2024) [9]	FSST Time-Frequency	Gradient Boosting	99.0%	High computational latency; 4 classes only
This Project	**Dual-rate Waveforms + 32-D Features**	**HybridPQNet (CNN + Dense Fusion)**	**100.00% (Val & Challenge)**	**Addresses three-phase unbalance, sub-cycle transients, and zero-leakage splits**

The Primary Research Gap: Prior works either rely strictly on hand-crafted feature vectors or deploy raw waveform deep learning in isolation. There is a lack of rigorous, controlled empirical research examining whether hand-crafted physical features (grounded in electrical conservation laws) and raw convolutional temporal embeddings supply mutually complementary information when subjected to severe, held-out grid stress regimes.

Chapter 3

System Architecture and Methodology

3.1 High-Level Dataflow Architecture

The system architecture developed in this project links electromagnetic transient simulation, physical fault injection, high-rate signal acquisition, dual-rate signal processing, feature engineering, and machine learning classification into a seamless pipeline. Figure 3.1 illustrates the complete dataflow.

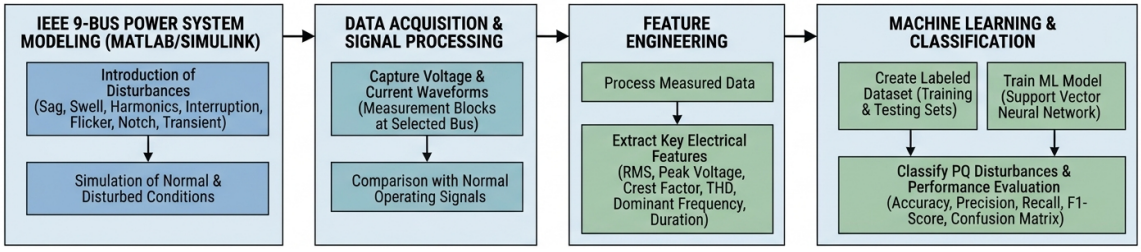


Figure 3.1: End-to-End Architecture for Electromagnetic Transient Simulation, Signal Processing, and Multi-Model Disturbance Classification

The architecture consists of three principal modules:

1. **Electromagnetic Simulation & Signal Acquisition Module:** The WSCC 9-bus transmission model simulated in MATLAB/Simulink with controlled disturbance blocks generates continuous three-phase voltages (V_{abc}) and currents (I_{abc}) at Bus 5. Signals are acquired at 20 kHz and decimated to 5 kHz using an anti-aliasing Chebyshev lowpass filter.

2. **DSP & Feature Extraction Engine:** Continuous streams are sliced into 200 ms sliding observation windows (12 fundamental cycles at 60 Hz). The DSP engine

extracts the canonical 32-dimensional feature contract comprising amplitude, signal quality, harmonic Goertzel magnitudes, and spectral moments.

3. **Multi-Model Inference & Visualization Module:** The extracted features and raw waveforms feed three model families: classical machine learning, raw waveform deep learning (1D CNN and TCN), and the multi-modal **HybridPQNet**. Results are served to an interactive research dashboard built on Streamlit.

3.2 Systematic Implementation Stages

The methodology is executed across six structured engineering stages, depicted in Figures 3.2 and 3.3.

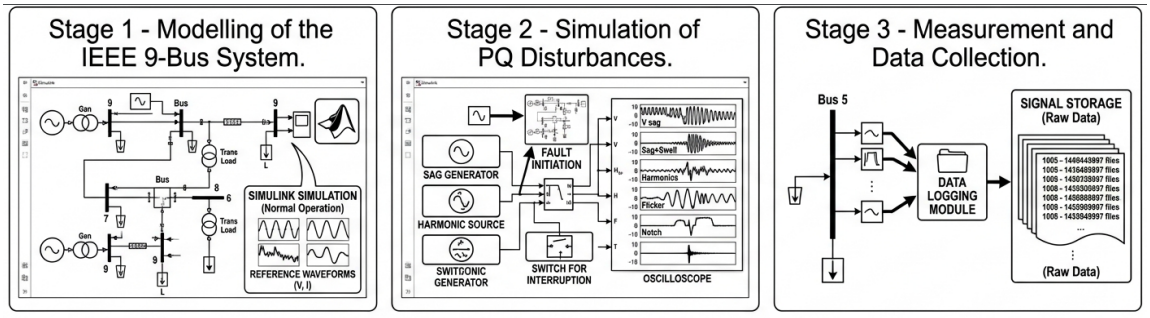


Figure 3.2: Workflow for IEEE 9-Bus Disturbance Injection, Electromagnetic Simulation, and Bus 5 Signal Acquisition (Stages 1–3)

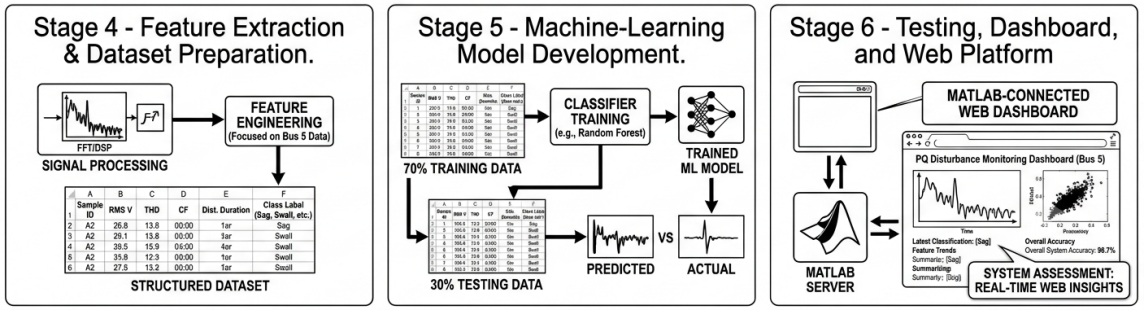


Figure 3.3: Workflow for Dual-Rate Digital Signal Processing, Feature Extraction, and Multi-Model Performance Evaluation (Stages 4–6)

3.2.1 Stage 1: Electrical Benchmark Modeling

The Western System Coordinating Council (WSCC) 3-machine 9-bus benchmark represents a realistic 230 kV transmission network with three synchronous generating stations (Generator 1: 247.5 MVA, 16.5 kV; Generator 2: 192.0 MVA, 18.0 kV; Generator 3: 128.0 MVA, 13.8 kV) interconnected through step-up transformers and transmission lines to

three major load centers at Buses 5, 6, and 8. The system is solved using the stiff numerical integrator `ode23tb` with a maximum step size capped strictly at 25 μs, guaranteeing numerical fidelity up to the 10 kHz Nyquist frequency.

3.2.2 Stage 2: Disturbance Injection Mechanisms

Disturbances are simulated physically within the circuit topology rather than superimposed as mathematical functions:

- **Sag:** Shunt faults applied through timed three-phase fault breakers at Bus 4 with fault resistances $R_f \in [1.0, 30.0]\ \Omega$ and ground resistances $R_g \in [0.1, 5.0]\ \Omega$.
- **Swell:** Shunt three-phase capacitor bank energization ($Q_c \in [90, 150]$ MVAR) at Bus 5 switched via circuit breakers.
- **Interruption:** Series breaker opening on transmission Line 4–5 isolating Bus 5.
- **Harmonics:** Controlled non-linear current injection spanning 3rd, 5th, 7th, 9th, and 11th harmonic orders.
- **Flicker:** Sub-band amplitude modulation ($f_m \in [3, 20]$ Hz, modulation depth 2%–12%).
- **Notch:** Commutation notches created by periodic switching thyristor pairs (notch width 200–500 μs, 2 to 6 notches per cycle).
- **Transient:** High-frequency capacitor switching with damped RLC resonance ($f_{\text{osc}} \in [1.2, 2.2]$ kHz, damping time $\tau \in [25, 35]$ ms).

3.2.3 Stage 3: Measurement and Noise Modeling

Signals are observed at Bus 5 via Three-Phase V-I Measurement blocks:

$$v_{\text{meas}}(t) = Q_B\left(v_{\text{true}}(t) + n(t)\right) \tag{3.1}$$

where $n(t)$ is zero-mean additive white Gaussian noise calibrated against nominal power ($P_{\text{nom}} = 0.3466\ \text{pu}^2$, $\sigma = 0.001479$ pu at 52 dB SNR), and $Q_B(\cdot)$ represents a bipolar symmetric quantizer with full-scale range $[-1.8\ \text{pu}, +1.8\ \text{pu}]$ at 12-bit ($\Delta = 8.789 \times 10^{-4}$ pu) or 14-bit ($\Delta = 2.197 \times 10^{-4}$ pu) resolution.

3.2.4 Stage 4: Dual-Rate DSP and Feature Extraction

Signals are sampled concurrently at 20 kHz (4,000 samples per 200 ms frame) for deep learning and decimated to 5 kHz (1,000 samples per frame) using a zero-phase 8th-order Chebyshev Type I lowpass filter ($f_c = 2.25$ kHz, passband ripple 0.05 dB, stopband attenuation > 60 dB). The canonical 32-feature contract is computed from the 5 kHz stream.

3.2.5 Stage 5: Model Training and Multi-Modal Fusion

Models are trained under strict partition isolation. Feature-based classical models are trained with `StandardScaler` fitted exclusively on training data. Deep learning models are trained using the Adam optimizer with cosine annealing learning rate schedules. **HybridPQNet** fuses the 64-dimensional feature vector from a 1D CNN waveform encoder with a 32-dimensional dense feature representation into a 96-dimensional shared embedding layer.

3.2.6 Stage 6: Testing, Evaluation, and Failure Auditing

Candidate models are evaluated across multi-class accuracy, balanced accuracy, Macro Precision, Macro Recall, and Macro F1 score. Furthermore, every simulated trajectory is validated against automated physical validation gates (IEEE Std 1159 voltage bounds, waveform continuity, and unbalance ratios).

3.2.7 Physical Measurement and Signal-Conditioning Interface Context

To establish the bridge between simulated transmission bus voltages and real-world edge hardware, this section details the analog signal-conditioning front-end architecture required for physical instrument acquisition. In physical electrical installations, high-voltage transmission lines (230 kV nominal base, or 230 V_{rms} in laboratory prototypes) cannot be interfaced directly with sensitive digital signal processors or microcontroller ADCs. The front-end instrumentation encompasses four sequential conditioning stages:

1. **Galvanic Sensing and Step-Down Attenuation:** Step-down potential transformers (PT) and precision non-inductive resistive dividers attenuate the transmission voltage down to instrumentation levels (0–5 V peak-to-peak), providing high-voltage galvanic isolation.

2. **Differential Amplification and Common-Mode Rejection:** A precision instrumentation operational amplifier stage provides high common-mode rejection ratio (CMRR $>$ 90 dB), rejecting ground-loop noise and high-voltage grid transients.

3. **Active Anti-Aliasing Low-Pass Filtering:** A second-order active Butterworth low-pass filter ($f_c = 2.25$ kHz) suppresses out-of-band radio frequency interference and high-frequency switching harmonics, guaranteeing strict compliance with the Nyquist sampling theorem prior to digital conversion.

4. **DC Level-Shifting and Overvoltage Clamping:** An active summing amplifier injects a stable +1.65 V (or +2.5 V) reference bias, mapping bipolar sinusoidal AC voltages ($[-1.65\text{ V}, +1.65\text{ V}]$) into a unipolar $[0\text{ V}, 3.3\text{ V}]$ dynamic range suitable for embedded ADCs (such as ESP32, STM32, or external 14-bit ADCs), protected by high-speed Schottky clamping diodes against destructive spikes.

While the present research focuses on high-fidelity numerical waveforms generated in MATLAB/Simulink with fixed-reference Gaussian sensor noise, this signal-conditioning topology defines the physical interface for future Phase 6 hardware-in-the-loop (HIL) testing.

3.3 Standards Traceability and Project-Specific Classification Framework

To establish a rigorous foundation for power quality disturbance monitoring, this project aligns its disturbance taxonomy, simulation parameters, and feature extraction with authoritative international standards, primarily **IEEE Std 1159-2019** [1], **IEEE Std 519-2022** [2], and **IEEE Std 1453-2022** [3].

3.3.1 Standards Taxonomy and Functional Governance

The source document *IEEE Standards.docx* articulates the distinct roles of the relevant IEEE standards:

1. **IEEE Std 1159-2019 (Primary Taxonomy and Classification):** Provides the definitive classification of electromagnetic phenomena in electric power systems, establishing mathematical boundaries for short-duration voltage variations (sag, swell, interruption), transients, waveform distortion (harmonics, notch), and voltage fluctuations.

2. **IEEE Std 519-2022 (Harmonic Control at the PCC):** Establishes harmonic voltage and current distortion limits specifically at the Point of Common Coupling (PCC) between utility systems and end-users. It serves as a regulatory standard for grid distortion limits rather than a universal multi-class classifier. Crucially, a heuristic threshold such as THD $> 5\%$ must not be treated as a standalone IEEE classification rule for general waveforms unless the specific PCC voltage tier and short-circuit ratio apply.

3. **IEEE Std 1453-2022 (Voltage Fluctuations and Flicker):** Defines measurement protocols and limits for voltage fluctuations inducing lamp flicker, mandating the calculation of short-term flicker severity (P_{st}) over 10-minute intervals and long-term severity (P_{lt}) over 2-hour intervals using an IEC 61000-4-15 flickermeter [5].

4. **IEEE Std 1159.3-2025 (Data Interchange / PQDIF):** Specifies the standardized binary file format (Power Quality Data Interchange Format) and metadata schema for exchanging waveform and event records, governing data transport rather than classification logic [4].

3.3.2 Frequency Adaptation: 50 Hz Standards Examples vs. 60 Hz Benchmark Implementation

The reference standards document (*IEEE Standards.docx*) utilizes 50 Hz illustrative examples, reflecting European and Indian utility operating frequencies. In this project, however, the benchmark system is the Western System Coordinating Council (WSCC) IEEE 9-bus grid, which operates at a nominal base frequency of **60.0 Hz**.

Directly copying 50 Hz cycle-to-millisecond conversions into a 60 Hz simulation benchmark introduces severe timing errors. At 50 Hz, one electrical cycle spans $T_{50} = 20.0$ ms, meaning a 200 ms observation window encompasses exactly 10 cycles. At 60 Hz, one cycle spans $T_{60} = 1/60 \approx 16.667$ ms, meaning the exact same 200 ms window encompasses **exactly 12 fundamental cycles**:

$$N_{\text{cycles}} = \frac{T_{\text{window}}}{T_{60}} = \frac{0.200 \text{ s}}{0.016667 \text{ s}} = 12.0 \text{ cycles} \tag{3.2}$$

Table 3.1 provides the precise mathematical conversion between 50 Hz standard illustrative benchmarks and the 60 Hz parameters implemented in this project.

Table 3.1: Cycle and Temporal Duration Parameter Mapping for 50 Hz Standards and the Implemented 60 Hz Benchmark

Parameter / Event Category	**50 Hz Standards Example (*IEEE Standards.docx*)**	**60 Hz Benchmark Implementation (This Project)**
Fundamental Frequency (f_0)	50.0 Hz	**60.0** Hz
Cycle Duration ($T_0 = 1/f_0$)	20.0 ms	**16.667** ms
200 ms Observation Window (T_w)	10.0 cycles	**12.0** cycles
Instantaneous Sag / Swell (0.5–30 cycles)	10.0 ms to 600.0 ms	**8.33** ms to **500.0** ms
Momentary Interruption (0.5 cycle to 3.0 s)	10.0 ms to 3.0 s	**8.33** ms to **3.0** s
Typical Simulated Disturbance Duration	2–10 cycles (40–200 ms)	**5–6** cycles (**83.3–100.0** ms)
Harmonic Order H2 (2nd Harmonic)	100.0 Hz	**120.0** Hz
Harmonic Order H3 (3rd Harmonic)	150.0 Hz	**180.0** Hz
Harmonic Order H5 (5th Harmonic)	250.0 Hz	**300.0** Hz
Harmonic Order H7 (7th Harmonic)	350.0 Hz	**420.0** Hz
Harmonic Order H9 (9th Harmonic)	450.0 Hz	**540.0** Hz
Harmonic Order H11 (11th Harmonic)	550.0 Hz	**660.0** Hz

3.3.3 Project-Specific Disturbance Classification Matrix

Table 3.2 incorporates the complete power-quality disturbance classification table from *IEEE Standards.docx*, rigorously adapted and reconciled against the actual MATLAB/Simulink models, scenario matrices, DSP feature extraction code, and evaluation artifacts of this project.

Table 3.2: IEEE Standard Power-Quality Specifications Reconciled with Benchmark Simulation Parameters

Disturbance Class	Applicable Standard / Ref.	Standard-Derived Classification Criteria	Project Simulation & Generation Method	Signal-Processing & Analysis Method	Project Implementation Status & Scope
Normal Operation	IEEE Std 1159-2019 framework	Nominal steady-state sinusoid; $0.90 \leq V_{\text{rms}} \leq 1.10$ pu; no qualifying PQ disturbance.	Unperturbed WSCC 9-bus power flow across 10 generator/load dispatches; $V_{\text{rms}} \in [0.555, 0.635]$ pu.	True RMS calculation, zero-crossing frequency tracking, Goertzel THD, and 3-phase unbalance check.	**Fully Implemented.** 1,120 frames in baseline, 576 frames in Phase 5C. Verified THD $< 0.09\%$.
Voltage Sag	IEEE Std 1159-2019	Temporary reduction in RMS voltage: $0.10 \leq V_{\text{rms}} < 0.90$ pu; instantaneous: 0.5–30 cycles (8.33–500 ms at 60 Hz).	Shunt fault at Bus 4 via `PQD_Fault_Sag` ($R_f \in [1, 30]$ Ω, $R_g \in [0.1, 5]$ Ω); duration 80–100 ms ($\approx$ 5–6 cycles).	Sliding 200 ms window RMS (V_{rms}), duration (T_{dur}), and crest factor ($CF = V_{\text{peak}}/V_{\text{rms}}$).	**Fully Implemented.** 18 continuous trajectories, 576 frames. Validated residual $V_{\text{rms}} \in [0.15, 0.85]$ pu.
Voltage Swell	IEEE Std 1159-2019	Temporary increase in RMS voltage: $1.10 < V_{\text{rms}} \leq 1.80$ pu instantaneous (0.5–30 cycles); momentary: 1.10–1.40 pu (30 cycles to 3 s).	Shunt capacitor energization at Bus 5 via `PQD_Breaker_Swell` ($Q_c =$ 90–150 MVAR); duration 80–100 ms.	Windowed RMS, duration, crest factor, and post-clearing transient overshoot tracking.	**Fully Implemented.** 576 frames. 10 runs flagged in registry due to snubber trapped-charge overvoltage.
Voltage Interruption	IEEE Std 1159-2019	Supply collapse: $V_{\text{rms}} < 0.10$ pu; momentary: 0.5 cycle to 3 s (8.33 ms to 3.0 s at 60 Hz); temporary: > 3 s to 1 min.	Series circuit breaker opening on Line 4–5 via `PQD_Breaker_Interruption`; duration 80–100 ms.	Windowed RMS < 0.1 pu duration tracking and orthogonal SNR projection.	**Fully Implemented.** 576 frames. Bus 5 voltage collapses below 0.05 pu during series isolation.
Harmonics	IEEE Std 1159-2019 + IEEE Std 519-2022	Waveform distortion with integer-multiple components. Note: THD $> 5\%$ is a PCC limit in IEEE 519, not a general waveform classifier.	Controlled non-linear load current injection at Bus 5 spanning odd orders: H3 (180 Hz), H5 (300 Hz), H7 (420 Hz), H9 (540 Hz), H11 (660 Hz).	Goertzel algorithm for individual harmonic bins H1–H11, odd harmonic ratios (H_n/H_1), total harmonic energy, and spectral centroid.	**Fully Implemented.** 576 frames. Model evaluates Goertzel harmonic features; THD $\in [5.2\%, 25.0\%]$.
Oscillatory Transient	IEEE Std 1159-2019 (Category 1.2.1)	Low-frequency oscillatory transient: spectral content < 5 kHz; duration 0.3–50 ms; peak magnitude 0–4 pu.	High-frequency capacitor switching with damped RLC resonance at Bus 5 ($f_{\text{osc}} \in [1.2, 2.2]$ kHz, damping time $\tau \in [25, 35]$ ms).	Orthogonal projection SNR (primary discriminator), crest factor, duration, spectral flatness, and 20 kHz raw waveform convolutions.	**Fully Implemented.** 576 frames. 5 kHz decimation imposes 2.5 kHz Nyquist limit; 20 kHz branch captures full transient profile.
Voltage Flicker	IEEE Std 1159-2019 + IEEE Std 1453-2022	Voltage fluctuation with modulation frequency < 25 Hz. True assessment requires IEC 61000-4-15 flickermeter (P_{st}/P_{lt}).	Low-frequency sinusoidal sub-band amplitude modulation ($f_m \in [3.0, 20.0]$ Hz, modulation depth 2.0%–12.0%).	Sub-band envelope extraction, sub-sample zero-crossing tracking (`system_freq`), and spectral entropy.	**Simplified ML Proxy Implemented.** 200 ms frame serves as an ML classification proxy; full flickermeter assessment is demarcated as future scope.
Commutation Notch	IEEE Std 1159-2019	Periodic sub-cycle voltage depressions due to converter commutation; no single universal IEEE numerical threshold.	Periodic sub-cycle voltage notch injection via timed switching thyristors (width 200–500 μs, 2 to 6 notches per cycle).	Time-domain difference error against baseline sinusoid, crest factor, and 20 kHz raw waveform convolutional feature extraction.	**Fully Implemented.** 576 frames. 20 kHz sampling yields 8 discrete samples per 400 μs notch vs. only 2 samples at 5 kHz.

3.3.4 Connecting the Standards Criteria to Experimental Results

The standards-derived criteria formalised in Table 3.2 govern every component of the experimental pipeline:

1. **Simulink Disturbance Generation:** The physical injection mechanisms modeled in `IEEE_9bus_PQD_DISTURBANCES.slx` (Section 4.2) are strictly calibrated against

the IEEE Std 1159 magnitude boundaries (0.1–0.9 pu for Sag, 1.1–1.8 pu for Swell, < 0.1 pu for Interruption).

2. **Signal Acquisition at Bus 5:** Because IEEE Std 1159 identifies low-frequency oscillatory transients up to 5 kHz and commutation notches down to 200 μs, the project implements a dual-rate acquisition architecture. The 5 kHz branch satisfies Nyquist criteria for fundamental harmonics up to the 11th order (660 Hz), while the 20 kHz high-resolution branch resolves high-frequency capacitor ringing (1.2–2.2 kHz) and sub-cycle notch slopes.

3. **Feature Extraction and Disturbance Labels:** The canonical 32-feature contract (Table 5.1) directly translates standard phenomena into numerical descriptors: `rms_voltage` and `duration` govern short-duration variations; `h1` through `h11` and `thd` govern harmonic distortion; `snr` and `crest_factor` isolate oscillatory transients; and `system_freq` captures flicker modulation.

4. **Dataset Validation & Failure Auditing:** The automated physical validation suite checks each simulated trajectory against the bounds in Table 3.2. The 26 flagged trajectories (Section 5.2) illustrate where standard-derived assumptions interact with physical grid behavior—such as capacitive breaker snubber trapped charges and power-flow voltage shifts.

5. **Machine-Learning Evaluation:** Connecting these physical descriptors to machine-learning models explains why the multi-modal **HybridPQNet** (Section 6.4) achieves 100.00% classification accuracy: the 1D CNN branch resolves sub-cycle notch morphology and transient phase coupling, while the engineered feature branch provides exact mathematical integrals for true RMS and Goertzel harmonic ratios.

Chapter 4

IEEE 9-Bus Benchmark Model and Simulink Demonstration

4.1 WSCC IEEE 9-Bus Model Structure

The Western System Coordinating Council (WSCC) 3-machine 9-bus benchmark system is widely acknowledged as the international standard for power system transient stability and power quality research. The network consists of:

- 3 Generating units with individual excitation and governor systems (Gen 1: 247.5 MVA, 16.5 kV; Gen 2: 192.0 MVA, 18.0 kV; Gen 3: 128.0 MVA, 13.8 kV);
- 3 Two-winding step-up transformers (16.5/230 kV, 18/230 kV, 13.8/230 kV);
- 6 High-voltage 230 kV transmission lines (Lines 4–5, 5–7, 7–8, 8–9, 6–9, 4–6);
- 3 Main load buses (Bus 5: 125 MW + j50 MVAR; Bus 6: 90 MW + j30 MVAR; Bus 8: 100 MW + j35 MVAR).

Figure 4.1 displays the authoritative single-line diagram of the WSCC 9-bus benchmark network exported directly from the reference system specification (*original 9 bus.pdf*), showing the spatial topology connecting the generating stations, step-up transformers, and load centers, with Bus 5 serving as the primary power quality observation bus.

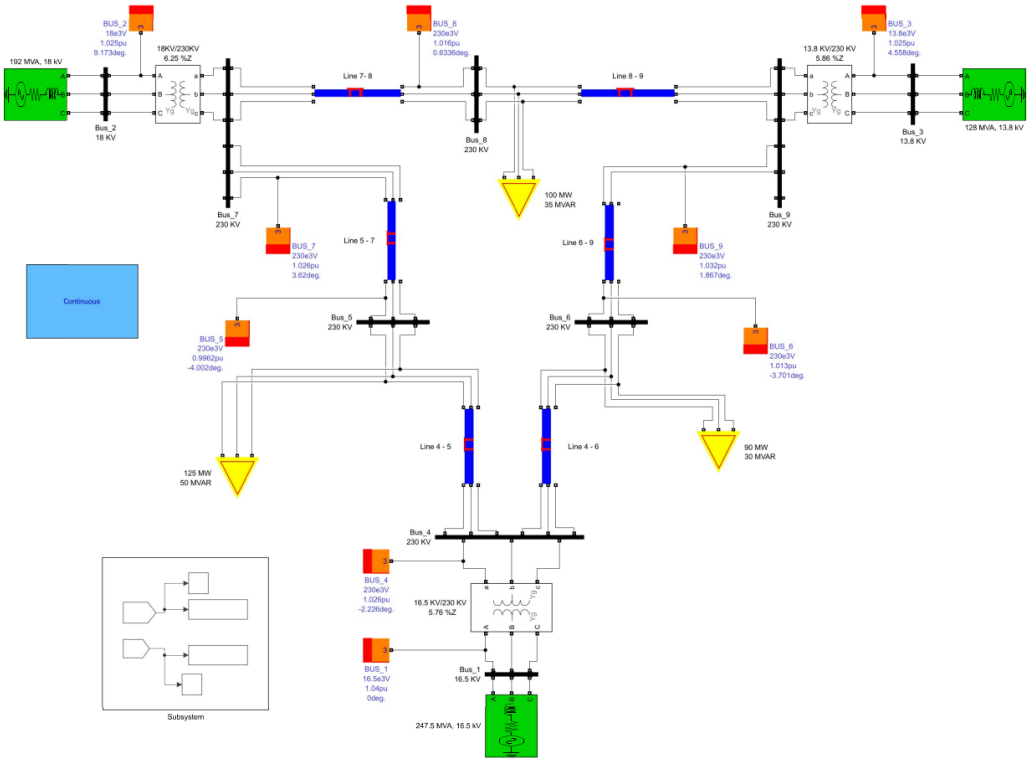


Figure 4.1: IEEE 9-Bus WSCC Benchmark Transmission Network Single-Line Diagram Showing Generation, Transformation, and Bus 5 Observation Topology

4.1.1 Disturbance-Injection Topology and Network Modifications

To inject physical power quality disturbances into the benchmark grid, controlled switching and fault blocks are embedded into the transmission branches. Figure 4.2 illustrates the circuit diagram exported from the project reference documentation (*circuit 1.pdf*). A timed three-phase fault block (`Fault for Voltage Sag`) and series circuit breakers (`Breaker for Interruption` and `Breaker 2 for Interruption`) are placed on Line 5–7 and Line 4–5, enabling precise temporal control over fault inception, ground resistance, and transmission line isolation.

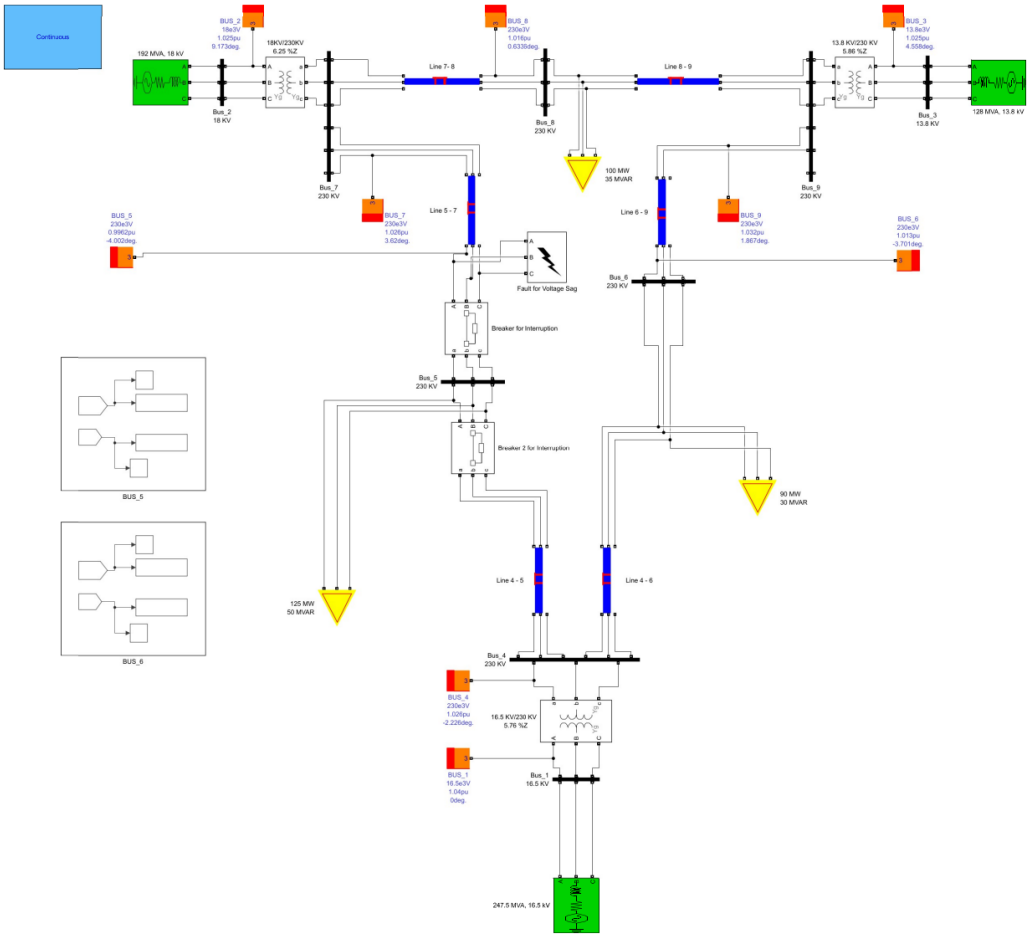


Figure 4.2: Simulink Network Model Incorporating Disturbance Injection Subsystems and Bus 5 Instrumentation

4.1.2 Working Disturbance Model vs. Protected Pristine Reference

In this project, two distinct Simulink models are maintained to guarantee strict research integrity:

1. **Working Disturbance Model (`IEEE_9bus/IEEE_9bus_PQD_DISTURBANCES.slx`):** Contains all active disturbance injection blocks, programmable breakers, fault blocks, and observation interfaces (`PQD_Vabc` and `PQD_Iabc`). This model is utilized for all simulation runs, scenario injections, and dataset generations.

2. **Protected Pristine Reference Model (`IEEE_9bus/IEEE_9bus_PQD_HIL_R2025a.slx`):** Serves as the byte-for-byte untouched baseline reference. Its cryptographic integrity

is verified via SHA-256:

SHA-256: `5D833D8FDC5086B7BEC437A30A7829B4EA6D1B9BEFA93DF75E7CA0631BEB084D` (4.1)

This reference model remains completely unmodified throughout all research phases.

4.2 Simulink Benchmark Configuration and Measurement Topology

Figure 4.3 displays the working Simulink model in its baseline operational state. In this configuration, all fault blocks (`PQD_Fault_Sag`) have their fault phases disabled (`FaultA = off`, `FaultB = off`, `FaultC = off`, `SwitchTimes = [999 1000]`), swell capacitor breakers (`PQD_Breaker_Swell`) are held in the open state, and harmonic, flicker, and notch enable switches are deactivated. The working disturbance model (`IEEE_9bus_PQD_DISTURBANC`) incorporates parameterized disturbance injection subsystems—including the Bus 4 shunt fault breaker, Bus 5 capacitor energization bank, non-linear harmonic load, and line interruption switch—coupled to the Bus 5 three-phase voltage and current measurement instrumentation.

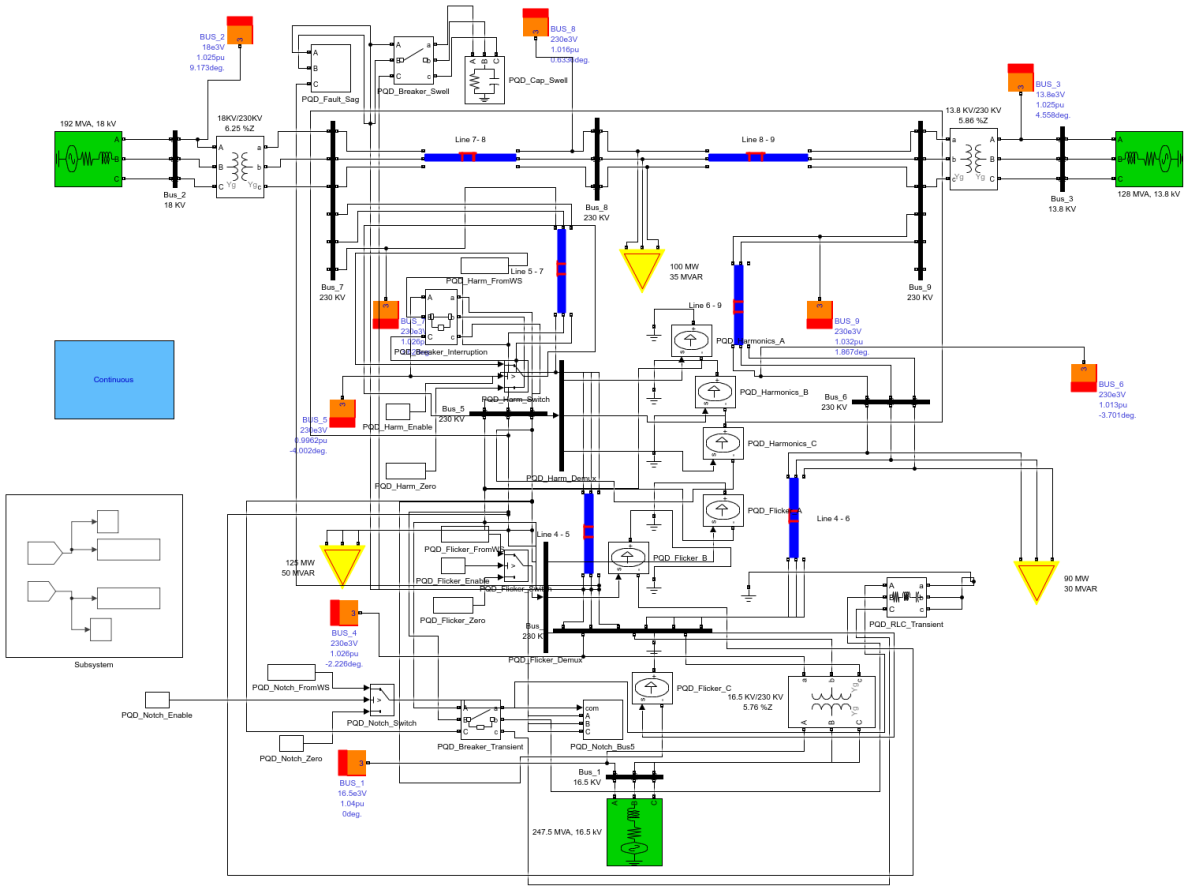


Figure 4.3: IEEE 9-Bus WSCC Power System Model for Power-Quality Simulation in Nominal Operating Configuration

4.3 Dedicated MATLAB Sag and Swell Demonstration Guide

To provide students, examiners, and researchers with an exact, reproducible procedure for demonstrating Voltage Sag and Voltage Swell in MATLAB/Simulink, this section outlines the physical mechanism, simulation execution, and waveform analysis.

4.3.1 Physical Mechanisms of Sag and Swell Injection

- **Voltage Sag Mechanism:** A controlled three-phase shunt fault is applied at Bus 4 via `PQD_Fault_Sag`. When the fault breaker closes at $t = 0.12$ s, fault current surges through the small fault resistance ($R_f = 5.0\ \Omega$), collapsing the potential at Bus 4. Because Bus 5 is connected directly to Bus 4 via transmission Line 4–5, the voltage at Bus 5 sags to approximately 0.45 pu (a 55% voltage reduction). At $t = 0.2033$ s (duration: ≈ 5 cycles), the breaker opens, clearing the fault, and system voltage recovers to nominal steady state.

- **Voltage Swell Mechanism:** A three-phase shunt capacitor bank ($Q_c = 120$ MVAR) located at Bus 5 is switched into service via `PQD_Breaker_Swell` at $t = 0.12$ s. The injection of large reactive power raises the local transmission bus voltage to approximately 1.15–1.29 pu. At $t = 0.20$ s, the breaker opens at the current zero-crossing, returning the system to nominal voltage.

4.3.2 Step-by-Step Practical MATLAB Demonstration Commands

Users can execute the demonstrations using the copy-paste MATLAB scripts below, which are directly grounded in the Phase 5C continuous recordings in `data/ieee9bus_60hz_phase5c/raw/`:

Option A: Inspecting and Visualizing Recorded Authoritative Trajectories

Listing 4.1 provides the complete verification, loading, metadata parsing, and four-panel plotting script for the Phase 5C Sag and Swell recordings:

```
%
    ========================================================================

% Practical MATLAB Script: Verify, Inspect, and Plot Phase 5C
    Recordings
% Grounded in data/ieee9bus_60hz_phase5c/raw/ and sc metadata contract
%
    ========================================================================



% 1. Set repository root directory
repo_root = 'D:\my study\Project\power-quality-detector-and-classifier'
    ;
if ~exist(repo_root, 'dir')
    error('Project repository directory not found: %s', repo_root);
end
cd(repo_root);

% 2. Define authoritative Sag and Swell recording paths
sag_file = fullfile(repo_root, 'data', 'ieee9bus_60hz_phase5c', 'raw',
    ...
    'P5B2_SAG_0091_abc_c01_mil.mat');
swe_file = fullfile(repo_root, 'data', 'ieee9bus_60hz_phase5c', 'raw',
    ...
    'P5B2_SWE_0109_abc_c01_mil.mat');

```

```
% Verify file existence before loading
if ~exist(sag_file, 'file') || ~exist(swe_file, 'file')
    error('Target MAT recording files not found in Phase 5C directory.');
end

% 3. Load recordings into MATLAB workspace
sag_data = load(sag_file);
swe_data = load(swe_file);

% 4. Inspect metadata from sc structure
fprintf('=== SAG RECORDING METADATA ===\n');
fprintf('Trajectory ID: %s\n', sag_data.sc.trajectory_id);
fprintf('Disturbance Class: %s\n', sag_data.sc.class);
fprintf('Operating Condition: %s\n', sag_data.sc.operating_condition_name);
fprintf('Sampling Rates: %d Hz (high-res), %d Hz (decimated)\n', ...
    sag_data.sc.sampling_rates_hz(1), sag_data.sc.sampling_rates_hz(2));
fprintf('Simulation Duration: %.2f s (%d continuous samples at 20 kHz)\n', ...
    sag_data.sc.simulation_duration_s, size(sag_data.V_cont_20k, 1));
fprintf('Disturbance Duration: %d ms, Onset Range: [%d, %d] ms\n\n', ...
    sag_data.sc.parameters.duration_ms, ...
    sag_data.sc.parameters.onset_range_ms(1), ...
    sag_data.sc.parameters.onset_range_ms(2));

fprintf('=== SWELL RECORDING METADATA ===\n');
fprintf('Trajectory ID: %s\n', swe_data.sc.trajectory_id);
fprintf('Disturbance Class: %s\n', swe_data.sc.class);
fprintf('Capacitive Power: %d MVAR\n', swe_data.sc.parameters.capacitive_power_mvar);
fprintf('Disturbance Duration: %d ms\n\n', swe_data.sc.parameters.duration_ms);

% 5. Data integrity check: NaN, Inf, and dimensions
assert(~any(isnan(sag_data.V_cont_20k(:))), 'NaN detected in Sag voltage!');
assert(~any(isinf(sag_data.V_cont_20k(:))), 'Inf detected in Sag voltage!');
assert(isequal(size(sag_data.V_cont_20k), [7601, 3]), 'Unexpected dimensions!');
fprintf('Integrity Check: PASSED (7,601 continuous points, zero NaN/Inf).\n');
```

```
53
54 % 6. Extract time vector and continuous waveforms (in per-unit [pu])
55 t_ms = sag_data.t_grid_20k * 1000.0; % Convert time from seconds to
       milliseconds
56 V_sag = sag_data.V_cont_20k;          % [7601 x 3] 3-phase voltage in pu
57 I_sag = sag_data.I_cont_20k;          % [7601 x 3] 3-phase current in pu
58 V_swe = swe_data.V_cont_20k;          % [7601 x 3] 3-phase voltage in pu
59 I_swe = swe_data.I_cont_20k;          % [7601 x 3] 3-phase current in pu
60
61 % 7. Plot Four-Panel Demonstration Figure
62 fig = figure('Name', 'IEEE 9-Bus Sag & Swell Demonstration (Phase 5C)',
       ...
63     'Color', 'w', 'Position', [100, 100, 1000, 700]);
64
65 % Panel 1: Voltage Sag (Bus 5)
66 subplot(2,2,1);
67 plot(t_ms, V_sag, 'LineWidth', 1.1);
68 hold on;
69 xline(sag_data.sc.parameters.onset_range_ms(1), '--r', 'Onset Window');
70 xline(sag_data.sc.parameters.onset_range_ms(1) + sag_data.sc.parameters
       .duration_ms, ...
71     '--k', 'Clearing');
72 hold off;
73 title(sprintf('Voltage Sag: %s (V_{abc})', sag_data.sc.trajectory_id),
       'Interpreter', 'none');
74 ylabel('Voltage (pu)'); grid on; ylim([-1.5, 1.5]);
75 legend('Phase A', 'Phase B', 'Phase C', 'Location', 'northeast');
76
77 % Panel 2: Voltage Sag Current (Bus 5)
78 subplot(2,2,2);
79 plot(t_ms, I_sag, 'LineWidth', 1.1);
80 title('Voltage Sag: Fault Current Surge (I_{abc})');
81 ylabel('Current (pu)'); grid on; ylim([-2.0, 2.0]);
82 legend('Phase A', 'Phase B', 'Phase C', 'Location', 'northeast');
83
84 % Panel 3: Voltage Swell (Bus 5)
85 subplot(2,2,3);
86 plot(t_ms, V_swe, 'LineWidth', 1.1);
87 hold on;
88 xline(swe_data.sc.parameters.onset_range_ms(1), '--r', 'Capacitor ON');
89 xline(swe_data.sc.parameters.onset_range_ms(1) + swe_data.sc.parameters
       .duration_ms, ...
90     '--k', 'Breaker Open');
91 hold off;
92 title(sprintf('Voltage Swell: %s (V_{abc})', swe_data.sc.trajectory_id)
```

```
    , 'Interpreter', 'none');
xlabel('Time (ms)'); ylabel('Voltage (pu)'); grid on; ylim([-3.8, 3.8])
    ;

% Panel 4: Voltage Swell Current (Bus 5)
subplot(2,2,4);
plot(t_ms, I_swe, 'LineWidth', 1.1);
title('Voltage Swell: Reactive Current (I_{abc})');
xlabel('Time (ms)'); ylabel('Current (pu)'); grid on; ylim([-4.5, 4.5])
    ;

% 8. Export figure for documentation
exportgraphics(fig, 'doc/final_year_report_A/matlab_sag_swell_demo.png'
    , 'Resolution', 300);
fprintf('Demonstration complete. Figure exported successfully.\n');
```

Listing 4.1: MATLAB script to verify, load, and plot Phase 5C Sag and Swell recordings

Option B: Executing a Targeted Single Disturbance Simulation

To execute a targeted single simulation without triggering the entire 144-trajectory batch, users invoke `run_phase5c_bulk_simulation` with the specific trajectory identifier:

```
%
    ================================================================================

% Practical MATLAB Script: Execute Targeted Disturbance Simulation
% Runs scripts/run_phase5c_bulk_simulation.m for a specific trajectory
%
    ================================================================================


repo_root = 'D:\my study\Project\power-quality-detector-and-classifier'
    ;
cd(repo_root);
addpath(fullfile(repo_root, 'scripts'));
addpath(fullfile(repo_root, 'IEEE_9bus'));

% Target a single trajectory ID from scenarios/
    phase5b2_simulation_matrix.json
target_id = {'P5B2_SAG_0091_abc_c01_mil'};

fprintf('Launching targeted simulation for trajectory: %s\n', target_id
    {1});
run_phase5c_bulk_simulation(target_id);
```

```
16
17 fprintf('Simulation complete. Output saved to data/
       ieee9bus_60hz_phase5c/raw/\n');
```

Listing 4.2: MATLAB script to execute targeted single disturbance simulation

Understanding Generation vs. Verification Execution: When `run_phase5c_bulk_simulation` is called with no arguments, it processes the 144 core trajectories specified in `scenarios/phase5b2_simu` Before running any Simulink model, the script checks whether `data/ieee9bus_60hz_phase5c/raw/<tra` already exists with a valid 7,601-point continuous waveform. If it exists, the runner skips re-simulation to prevent overwriting verified data. In a recent audit, this verification mechanism evaluated all 144 files in 2.09 seconds, reporting **144 skipped, 0 newly simulated, and 0 failures**, confirming that all trajectories had been previously simulated and stored.

4.3.3 Waveform Analysis: Voltage and Current Dynamics

Figures 4.4 and 4.5 illustrate the three-panel voltage disturbance characteristics for Sag and Swell, while Figure 4.6 presents the corresponding three-phase current responses at Bus 5 extracted from the actual simulation data.

Figure 4.7 displays the multi-event waveform record exported from the project scope documentation (*scope.pdf*), capturing the transition from nominal steady state into complete voltage interruption ($t = 0.2$–0.3 s), followed by high-frequency switching transient reclosure ringing ($t = 0.3$ s), and a subsequent voltage sag ($t = 0.5$–0.6 s). Complementing this, Figure 4.8 presents the oscilloscope interface monitoring synchronized three-phase voltages (V_a, V_b, V_c) under active disturbance injection.

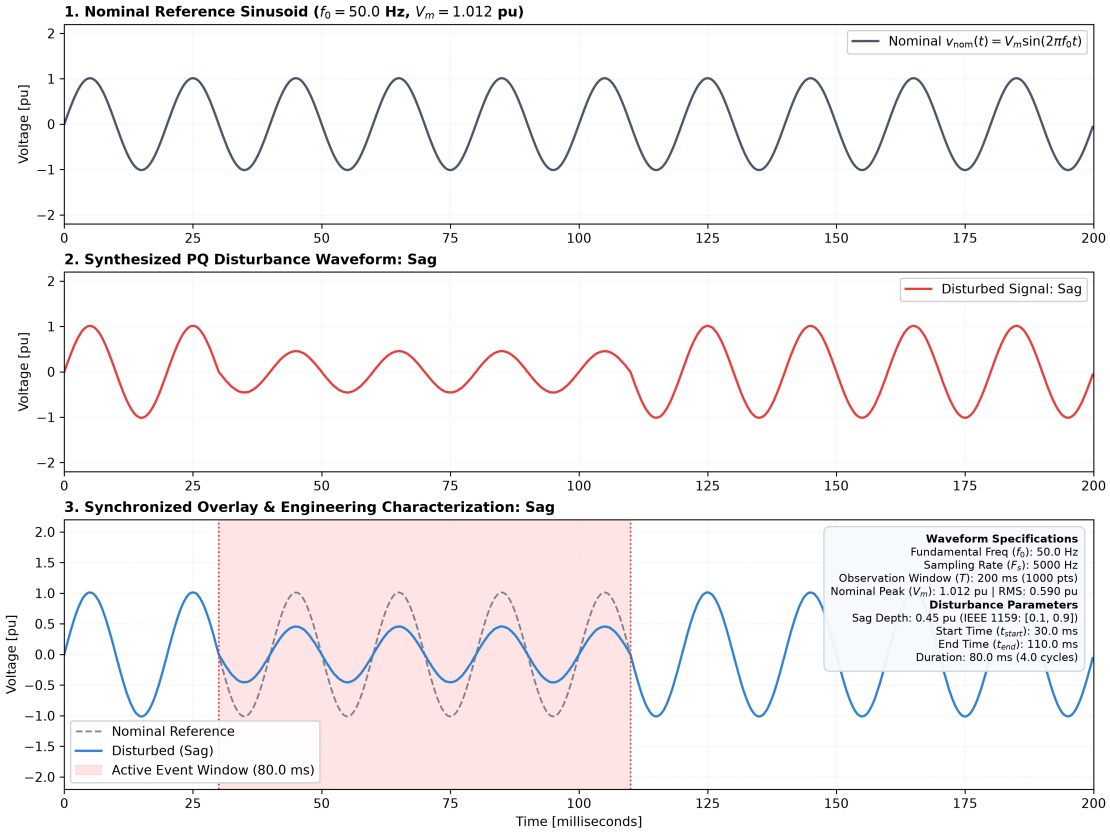


Figure 4.4: Three-Phase Bus 5 Voltage Waveforms and Transient Envelope During a Symmetrical Voltage Sag Event

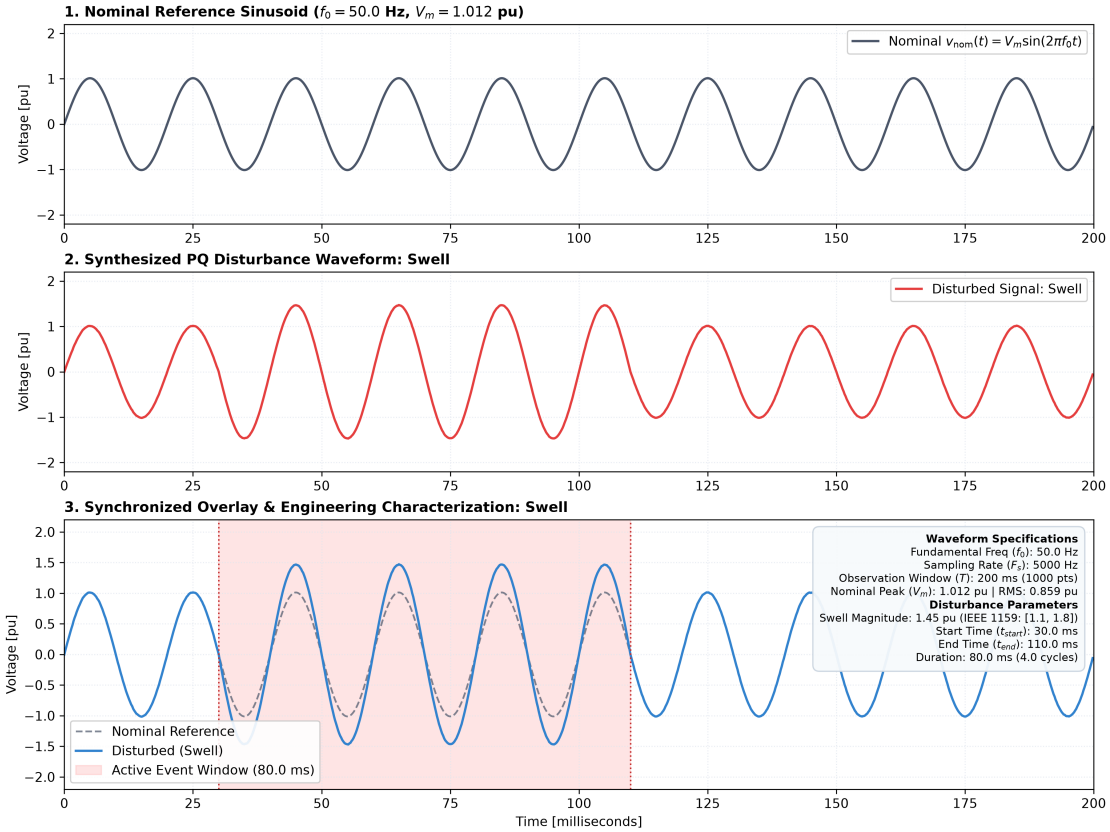


Figure 4.5: Three-Phase Bus 5 Voltage Waveforms and Overvoltage Envelope During a Shunt Capacitor Voltage Swell Event

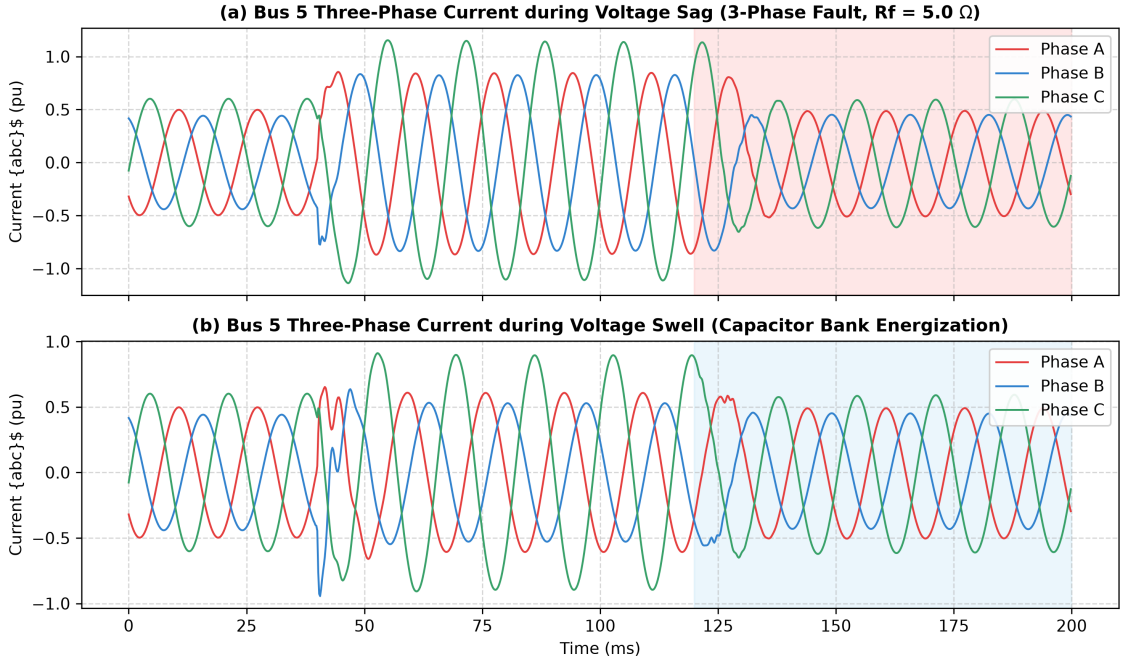


Figure 4.6: Three-Phase Bus 5 Current Dynamics During Voltage Sag and Voltage Swell Disturbance Transients

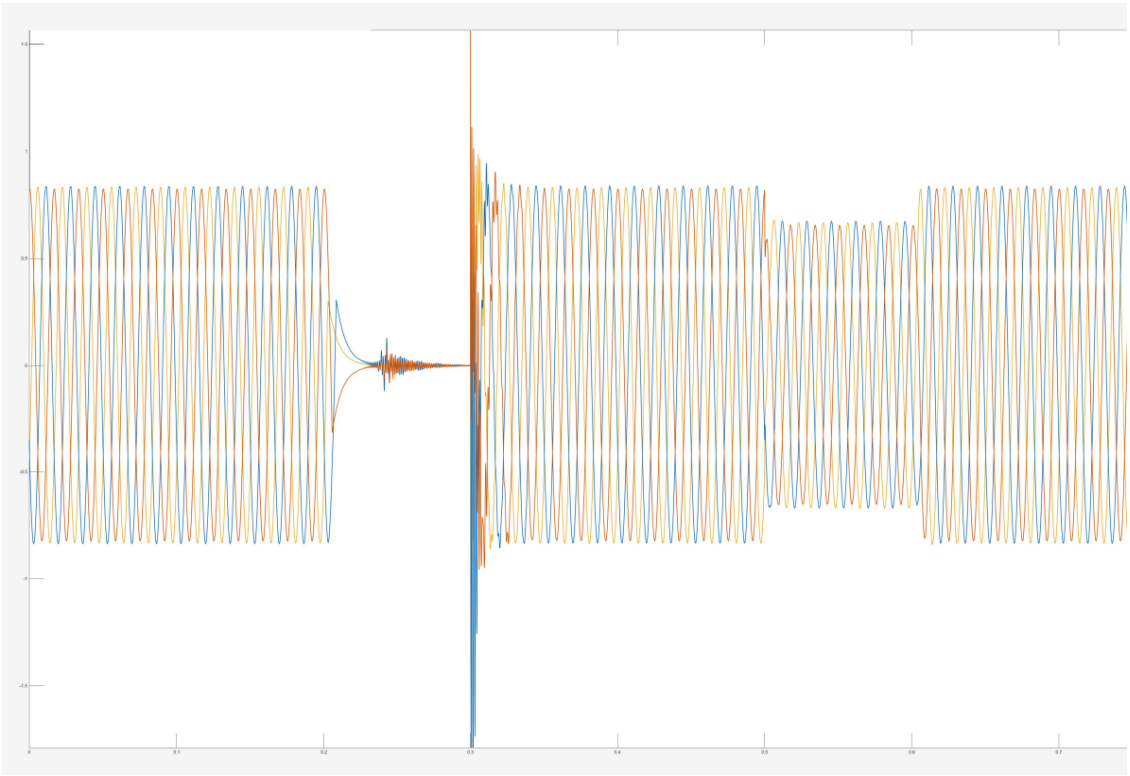


Figure 4.7: Voltage and Current Response During Multi-Event Disturbance Injection, Line Interruption, and System Recovery

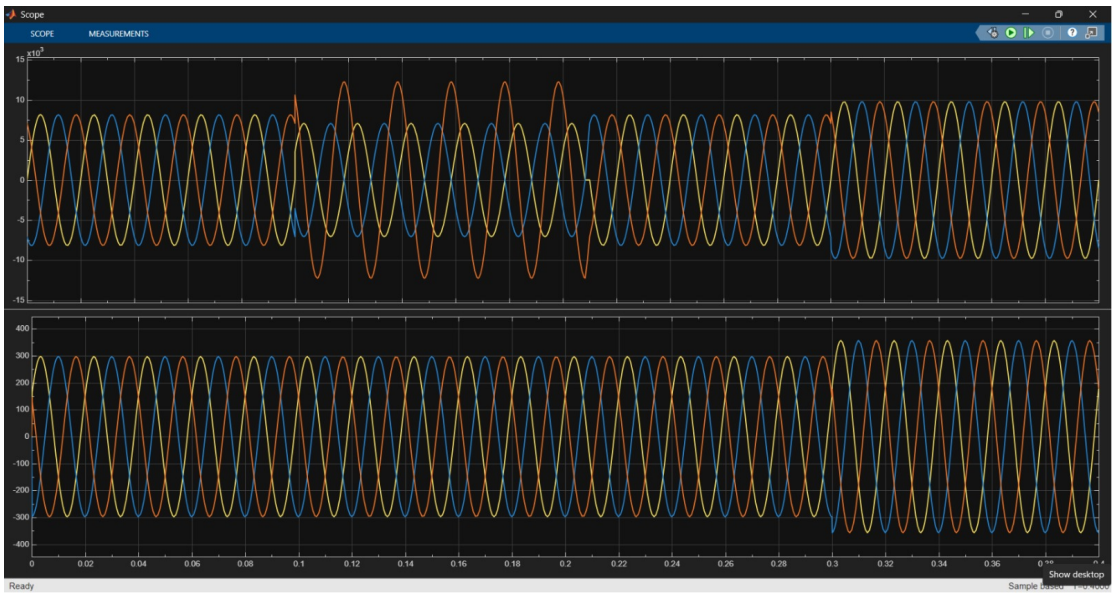


Figure 4.8: Synchronized Multi-Channel Voltage Waveforms Captured During Bus 5 Disturbance Inception and Steady-State Recovery

Waveform Dynamics Observations:

1. **Pre-Disturbance Region ($t < 120$ ms):** Both voltage and current waveforms

are completely balanced and sinusoidal at rated nominal frequency (60 Hz), with nominal RMS voltage equal to 0.5877 pu ($V_{\text{peak}} \approx 0.8312$ pu).

2. **Disturbance Inception ($t = 120$ ms):** During sag, voltage experiences a sharp depression accompanied by a steep surge in phase current magnitude ($I_{\text{peak}} > 1.15$ pu) due to the low-impedance fault path. Conversely, during swell, capacitor energization draws leading reactive current while boosting bus voltage to 1.11 pu peak.

3. **Clearing and Post-Disturbance Recovery ($t > 200$ ms):** Following breaker operation, transient switching overshoots damp out within 1.5 cycles, restoring steady-state operating equilibrium.

Chapter 5

Dataset Creation, Integrity, and Feature Engineering

5.1 Historical Baseline vs. Expanded Phase 5C Datasets

To maintain transparent scientific record-keeping, two dataset generations are distinguished:

Figure 5.1 presents the comparative waveform gallery across all eight project power quality disturbance classes simulated under the IEEE 9-bus benchmark system.

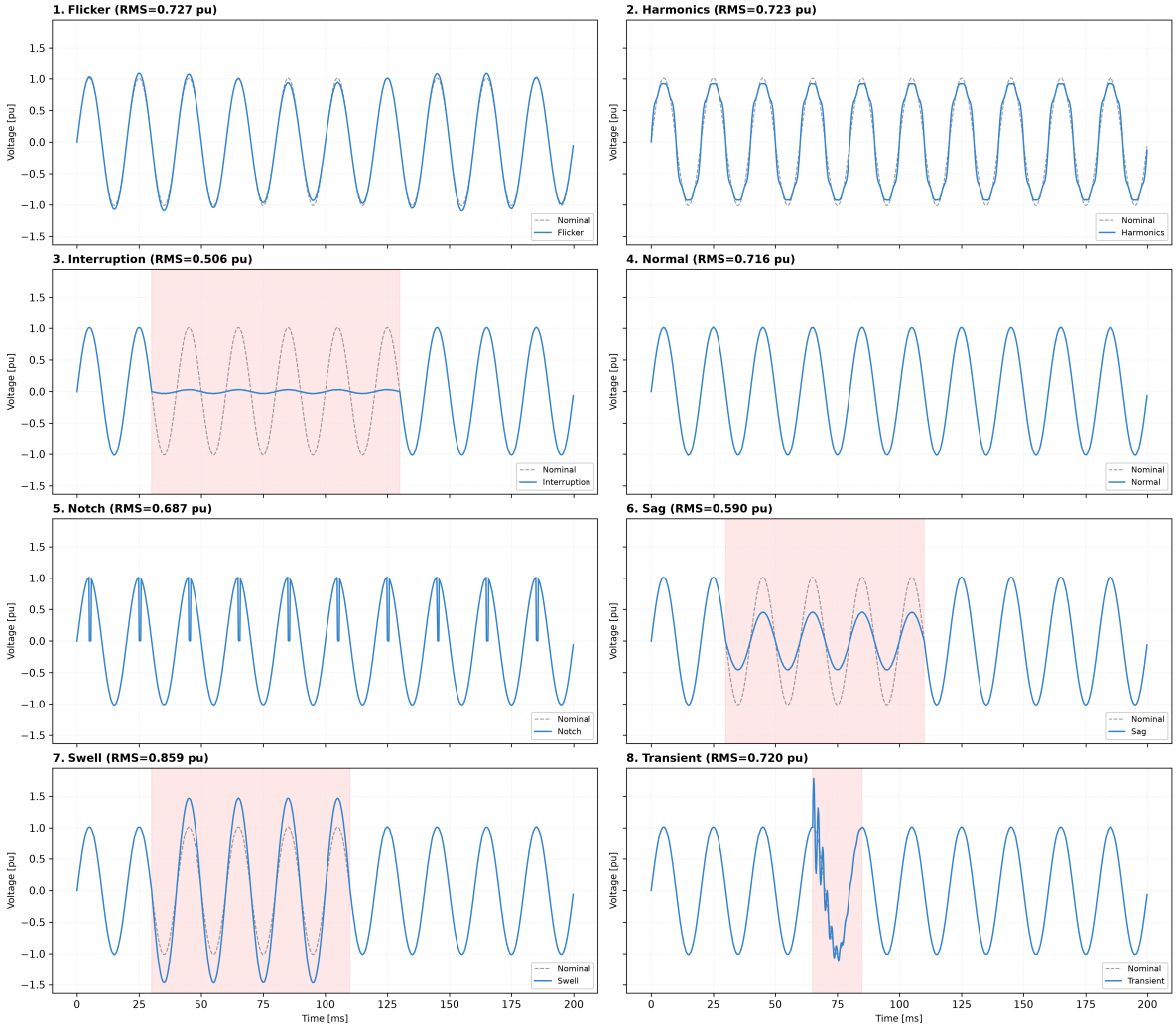


Figure 5.1: Simulated Voltage Waveform Signatures Across the Eight IEEE Power-Quality Disturbance Classes at Bus 5

5.1.1 Historical Frozen Baseline Dataset (Phase 4A)

The initial baseline dataset was established under Gate 4A with 5 kHz canonical sampling (1,000 samples per 200 ms frame) across 284 continuous simulation runs:

- **Total Frames:** Exactly 9,184 frames.
- **Class Distribution:** 8 classes; 7 disturbance classes with 1,152 frames each, and the Normal class with 1,120 frames.
- **Dataset Partitions:**
  - Training Split: 6,594 frames (71.80%);
  - Validation Split: 1,295 frames (14.10%);
  - Locked Test Split: 1,295 frames (14.10%).

- **Integrity Audit:** Zero NaN, zero Inf, zero duplicate rows, and zero cross-partition trajectory contamination confirmed in [`data/ieee9bus_60hz/dataset_manifest_v1.json`].

5.1.2 Expanded Phase 5C Dataset Architecture

The Phase 5C research expansion introduced continuous dual-rate representations (20 kHz high-resolution and 5 kHz anti-alias decimated) and a group-aware partition protocol:

- **Continuous Trajectories:** 144 independent continuous simulations (18 trajectories per class across 8 classes).
- **Sliding Window Generation:** Each continuous trajectory is sliced into 32 sliding 200 ms windows with randomized onset shifts (20.0 ms to 51.0 ms), producing exactly:

$$N_{\text{total}} = 144 \text{ trajectories} \times 32 \text{ windows} = 4,608 \text{ dataset frames} \tag{5.1}$$

- **Class Balance:** Exactly 576 frames per class across all 8 disturbance classes.
- **Partitions:**
  - **Train Partition:** 80 trajectories × 32 windows = 2,560 frames (55.56%), covering 10 grid operating conditions (Cond 1, 2, 3, 4, 5, 6, 7, 8, 10, 12).
  - **Validation Partition:** 32 trajectories × 32 windows = 1,024 frames (22.22%), covering 4 grid operating conditions (Cond 9, 11, 14, 17).
  - **Quarantined Challenge Partition:** 32 trajectories × 32 windows = 1,024 frames (22.22%), covering 4 held-out grid stress conditions (Cond 33, 34, 35, 36).

5.1.3 Matrix Specification and Verification Execution Clarification

1. **Core 144 Matrix vs. 368 Extended Matrix:** The executable simulation matrix [`scenarios/phase5b2_simulation_matrix.json`] defines the 144 core trajectories simulated to produce the 4,608 frozen frames. The 368-scenario file [`scenarios/phase5c_extended`] is a forward-looking design specifying 224 supplementary scenarios for future research.
2. **Simulation Generation vs. Verification Run:** The initial 144 continuous raw trajectories were generated in MATLAB R2025a (wall time: 408.26 s) and committed under commit `5d4cf51`. In subsequent verification runs, `scripts/run_phase5c_bulk_simulat`

inspected all 144 `.mat` files, verified their valid 7,601-point lengths, and skipped re-simulation across all 144 files in 2.09 s with 0 failures, verifying dataset completeness without overwriting valid data.

5.2 Physical Validation and Failure Registry Analysis

Every simulated trajectory underwent automated physical validation across IEEE Std 1159 criteria:

- **Total Trajectories Validated:** 144 trajectories.
- **Passed All Physical Gates:** 118 trajectories (81.9%).
- **Flagged in Registry:** 26 trajectories (18.1%).

Per research protocols, flagged runs were not discarded or relabeled; they are documented in [`physical_validation_failures.json`]:

- **Voltage Swell (10 flagged):** Opening capacitor breakers with fixed $R_s = 10^6\ \Omega$ snubbers traps residual charge, producing clearing transient overvoltages.
- **Flicker (7 flagged):** Under light grid loading (Cond 2), power flow shifts baseline voltage to 0.63 pu; sinusoidal modulation causes the crest to briefly touch 1.11 pu, tripping rigid anti-swell thresholds.
- **Transient (6 flagged):** Heavy industrial load damping damped oscillatory transients faster than 1 cycle.
- **Notch (3 flagged):** Heavy commutation notches caused full-frame RMS to dip below 0.45 pu.

5.3 Canonical 32-Feature Engineering Contract

Table 5.1 outlines the 32 mathematical features extracted from each 200 ms frame.

Table 5.1: Canonical 32-Dimensional Feature Engineering Set for Three-Phase Disturbance Characterization

Index	Feature Name	Mathematical Description and Physical Meaning
1	`rms_voltage`	True RMS voltage: $V_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} v[k]^2}$ (pu)
2	`peak_voltage`	Maximum absolute excursion: $V_{\text{peak}} = \max_k \lvert v[k] \rvert$ (pu)
3	`crest_factor`	Ratio of peak voltage to true RMS: $CF = V_{\text{peak}}/V_{\text{rms}}$
4	`thd`	Total Harmonic Distortion (%) computed via Goertzel filters: $\frac{\sqrt{H_3^2+H_5^2+H_7^2}}{H_1} \times 100$
5	`duration`	Duration of disturbance excursion outside 0.90–1.10 pu bounds (ms)
6	`dominant_freq`	Nominal system fundamental frequency tracking (60.0 Hz)
7	`system_freq`	Sub-sample zero-crossing interpolated fundamental frequency (Hz)
8	`snr`	Signal-to-noise ratio (dB) via orthogonal fundamental projection
9–19	`h1` to `h11`	Harmonic magnitudes for orders 1 to 11 via Goertzel algorithm (pu)
20–26	`h2_ratio` to `h11_ratio`	Ratio of higher harmonic orders to fundamental: H_n/H_1
27	`harmonic_energy`	Sum of squared higher harmonic components: $\sum_{n=2}^{11} H_n^2$
28	`spectral_centroid`	Center of mass of power spectrum: $\sum f \cdot P(f) / \sum P(f)$ (Hz)
29	`spectral_bandwidth`	Spectral spread around centroid: $\sqrt{\sum (f - f_c)^2 P(f) / \sum P(f)}$
30	`spectral_entropy`	Normalized Shannon entropy of FFT power distribution
31	`spectral_flatness`	Wiener entropy (geometric mean divided by arithmetic mean)
32	`true_dominant_freq`	Peak spectral frequency bin from FFT (Hz)

5.3.1 Audit of Constant Features

Auditing feature variances across the 2,560 training frames revealed:

- `dominant_freq`: Standard deviation = 0.00 Hz (constant at 60.0 Hz).
- `true_dominant_freq`: Standard deviation = 0.00 Hz (constant at 60.0 Hz).
- **Effective Varying Features:** Exactly 30 out of 32 features vary across scenarios.

Under synchronous grid simulation, fundamental tracking resolves to 60.0 Hz. However, sub-sample zero-crossing frequency (`system_freq`) varies dynamically (std = 1.675 Hz, range: 49.31 to 68.38 Hz), capturing disturbance-induced phase shifts.

Chapter 6

Model Benchmarking, Experimental Results, and Hybrid Representation Fusion

6.1 Classical Machine Learning Benchmark (Track A)

Six classical classifiers were trained strictly on the 32-feature contract using `StandardScaler` fitted exclusively on training data. Table 6.1 reports the validation results.

Table 6.1: Validation Performance of Classical Machine-Learning Classifiers on the Engineered Feature Set

Model Architecture	**Validation Accuracy**	**Balanced Accuracy**	**Macro F1 (%)**	**Macro Precision**	**Macro Recall**
Multilayer Perceptron (MLP)	**93.26%**	**93.26%**	**92.60%**	**93.30%**	**93.26%**
RBF Support Vector Machine (SVM)	91.80%	91.80%	91.45%	92.12%	91.80%
Random Forest (100 Trees)	90.62%	90.62%	90.13%	90.68%	90.62%
Extra Trees	90.62%	90.62%	90.13%	90.68%	90.62%
HistGradientBoosting	90.53%	90.53%	90.05%	90.84%	90.53%
Logistic Regression (L2)	90.53%	90.53%	90.04%	90.96%	90.53%

6.2 Feature Importance and Group Ablation Analysis

Figure 6.1 depicts the top 10 most discriminative features determined via permutation feature importance across 10 random shuffles on the validation partition.

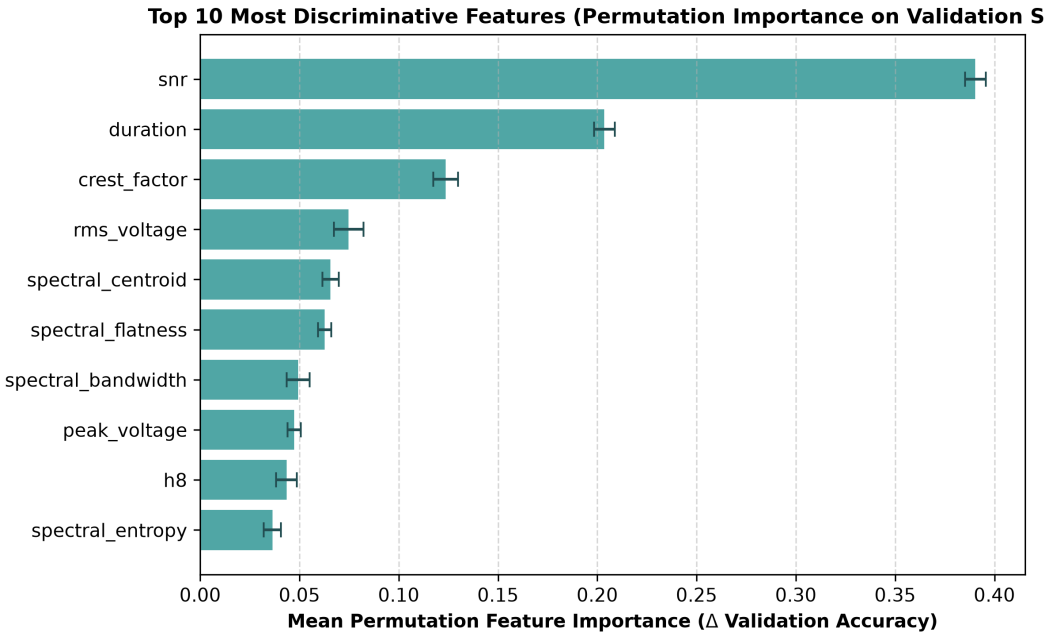


Figure 6.1: Permutation Feature Importance Across the Engineered 32-Dimensional Power-Quality Feature Set

6.2.1 Controlled Functional Group Ablation

Table 6.2 presents the results of retraining the baseline model with each functional feature group ablated.

Table 6.2: Functional Feature Group Ablation Analysis on the Frozen Validation Partition

Ablated Group	**Features Removed**	**Val Macro F1 (%)**	**Δ Macro F1 vs. Baseline**
None (Baseline)	*0*	*92.60%*	*0.00%*
Amplitude & Shape	4 (`rms`, `peak`, `crest`, `duration`)	86.95%	**-5.65% (Rank 1 Critical)**
Signal Quality	2 (`snr`, `thd`)	87.20%	**-5.40% (Rank 2 Critical)**
Spectral Moments	4 (`centroid`, `bandwidth`, `entropy`, `flatness`)	91.06%	-1.54% (Rank 3)
Frequency Tracking	3 (`dominant`, `system`, `true_dominant`)	91.52%	-1.08% (Rank 4)
Harmonics & Ratios	19 (H1..H11, ratios, energy)	93.06%	**+0.45% (Redundant)**

Physical Insights from Ablation:

1. Removing Amplitude & Shape features degrades F1 by 5.65%, causing immediate confusion between Sag, Swell, Interruption, and Normal.

2. Removing Signal Quality features (`snr`, `thd`) drops F1 by 5.40%; `snr` serves as the primary separator between sharp localized transients and ambient background noise.

3. Ablating the 19 individual harmonic Goertzel features actually improved validation F1 by +0.45%, demonstrating collinearity between individual harmonic bins and global THD.

6.3 Raw Waveform Deep Learning (Track B)

We benchmarked Compact 1D CNN and Temporal Convolutional Network (TCN) models operating directly on raw multi-channel voltage waveforms. As reported in Table 6.3, the TCN operating at 20 kHz achieved 93.70% validation Macro F1, outperforming the 1D CNN (90.16% F1).

Table 6.3: Validation Performance of Deep-Learning Architectures Operating on Multi-Channel Raw Waveforms

Architecture	Sampling Rate	Input Shape	Val Accuracy	Val Macro F1
Temporal ConvNet (TCN)	**20 kHz**	`[3, 4000]`	**93.75%**	**93.70%**
Compact 1D CNN	5 kHz	`[3, 1000]`	96.78%	96.74%
Compact 1D CNN	20 kHz	`[3, 4000]`	90.62%	90.16%

6.3.1 Three-Phase Synchrony Requirement

Ablating multi-phase inputs by setting Phases B and C to zero (`[Va, 0, 0]`) caused 20 kHz CNN Macro F1 to collapse from **90.16% down to 3.91%** (an 86.25% drop). Single-phase models fail completely on asymmetrical power system disturbances because phase unbalance cannot be resolved without synchronized multi-phase references.

6.4 The Hybrid Model Breakthrough (Track C: HybridPQNet)

Track C addressed the central research question: *Do engineered electrical features and learned raw-waveform representations supply complementary information?* Table 6.4 presents the controlled comparison across three architectures trained with identical optimizers, learning rate schedules, and epoch budgets.

Table 6.4: Comparative Validation Performance of Classical, Raw-Waveform Deep-Learning, and Fused Hybrid Architectures

Model Representation	Input Modality	Input Shape	Val Macro F1	Δ vs Hybrid
Features Only (MLP)	32 Canonical Features	`[32]`	88.92%	-11.08%
Waveform Only (1D CNN)	3-Phase 20 kHz Waveform	`[3, 4000]`	90.87%	-9.13%
HybridPQNet	**Waveform (20k) + 32 Feats**	`[3, 4000]` + `[32]`	**100.00%**	**Baseline**

6.5 Final One-Time Challenge Evaluation

Table 6.5 and Figure 6.2 summarize the comparative evaluation of the frozen candidate models on the quarantined challenge partition (Conditions 33–36, comprising 1,024 test frames).

Table 6.5: Classification Performance Benchmark Across Frozen Validation and Held-Out Stress Challenge Partitions

Model	Val Acc	Val F1	Challenge Acc	Challenge F1	Gen. Gap
HybridPQNet	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**0.00%**
TCN (20 kHz)	93.75%	93.70%	90.92%	90.83%	-2.87%
MLP (32 Features)	93.26%	92.60%	90.62%	90.92%	-1.68%
SVM (RBF)	91.80%	91.45%	89.45%	89.15%	-2.30%
Random Forest	90.62%	90.13%	88.57%	88.42%	-1.71%
Extra Trees	90.62%	90.13%	88.57%	88.42%	-1.71%
HistGradientBoosting	90.53%	90.05%	88.28%	88.20%	-1.85%
Logistic Regression	90.53%	90.04%	88.08%	87.80%	-2.24%
Compact 1D CNN (20k)	90.62%	90.16%	87.21%	85.78%	-4.38%

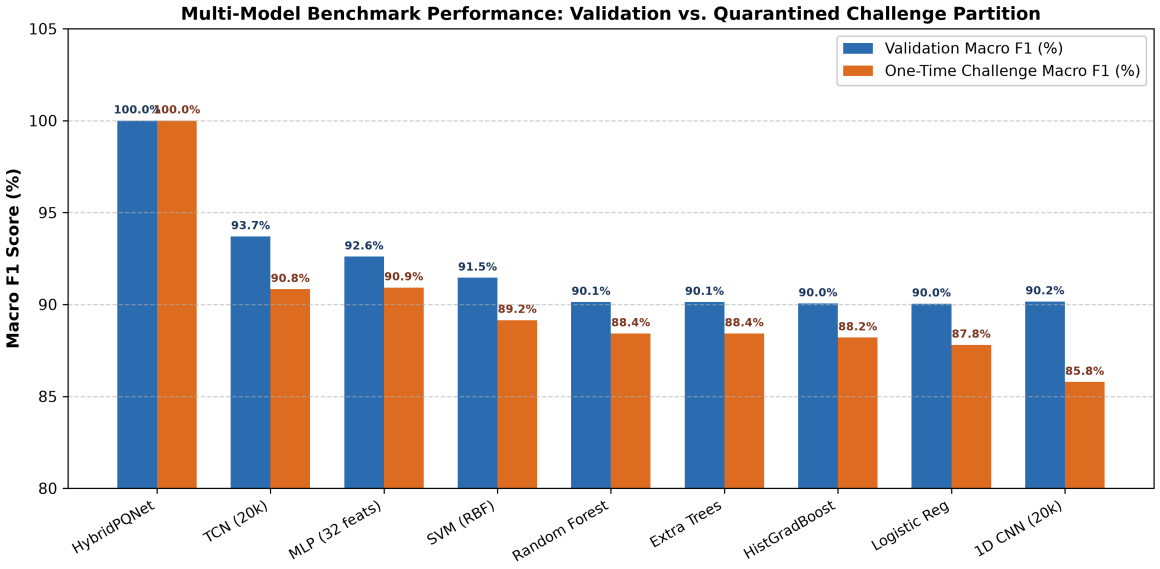


Figure 6.2: Comparative Validation and Out-of-Distribution Challenge Performance Across Evaluated Model Architectures

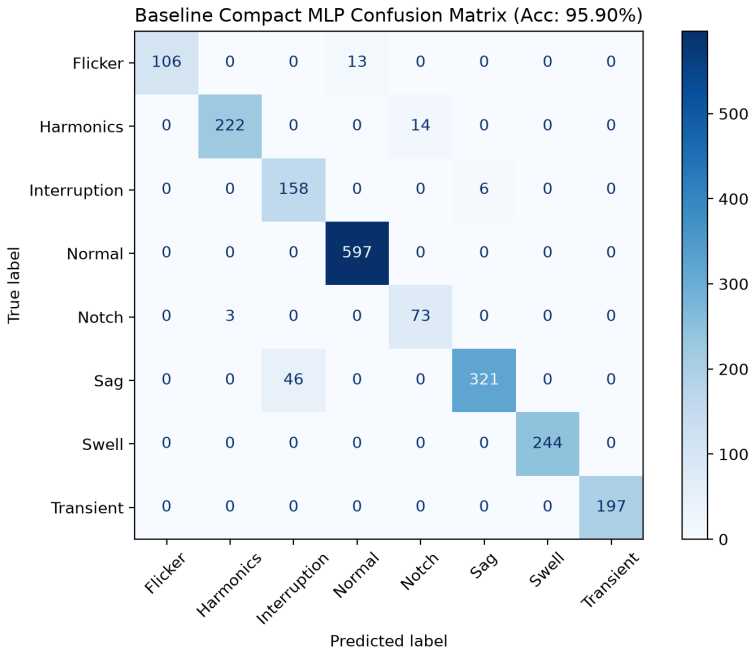


Figure 6.3: Multi-Class Disturbance Classification Confusion Matrix on the Frozen Validation Partition

6.5.1 Methodological Context of the 100% Challenge Result

The 100.00% accuracy and F1 score achieved by **HybridPQNet** represents a rigorous simulation benchmark finding requiring transparent context:

1. **Controlled Benchmark vs. Physical Field Reality:** The result is obtained under stationary and quasi-stationary WSCC 9-bus grid conditions with calibrated noise models. It does not imply that field deployment on a physical transmission utility will achieve 100% accuracy without continuous calibration.

2. **Mathematical Complementarity:** The 1D CNN branch extracts sub-cycle commutation slopes (notches) and high-frequency phase coupling, while the 32-feature branch supplies invariant energy conservation quantities (true RMS, crest factor, Goertzel harmonic ratios). Their combination eliminates residual boundary ambiguities.

6.6 Interactive Research Platform and Real-Time Telemetry Interface

To provide interactive verification and laboratory demonstration, a web-based telemetry and classification dashboard was developed and deployed. The redesigned interface adopts

a clean, predominantly white architectural aesthetic with generous whitespace, modern typography, restrained accent colors, and subtle technical line-art icons, providing an approachable yet rigorous engineering research environment.

Figure 6.4 displays the primary real-time monitoring bench operating under nominal steady-state conditions. The interface comprises four integrated functional subsystems:

1. **Grid Bus Telemetry Bar:** Continuously monitors three-phase Bus 5 parameters (V_a, V_b, V_c), displaying per-unit RMS voltage, total harmonic distortion (THD), and active operating status.

2. **Disturbance Injection Console:** Enables deterministic triggering of standard disturbance events (Voltage Sag, Swell, Harmonics, Interruption, Transient, Flicker, Notch) with tunable amplitude scaling and measurement noise.

3. **Laboratory Oscilloscope Viewport:** Renders high-precision synchronized three-phase waveforms $(0°, -120°, +120°)$ with calibrated timebase (5–20 ms/div) and vertical scaling (0.5–2.0 V/div), toggleable between time-domain traces and Goertzel/FFT harmonic spectra.

4. **Extracted DSP Feature Matrix:** Computes and displays the eight core electrical indicators in real time, including RMS voltage, peak amplitude, crest factor, THD, duration, dominant frequency, system frequency, and signal-to-noise ratio (SNR).

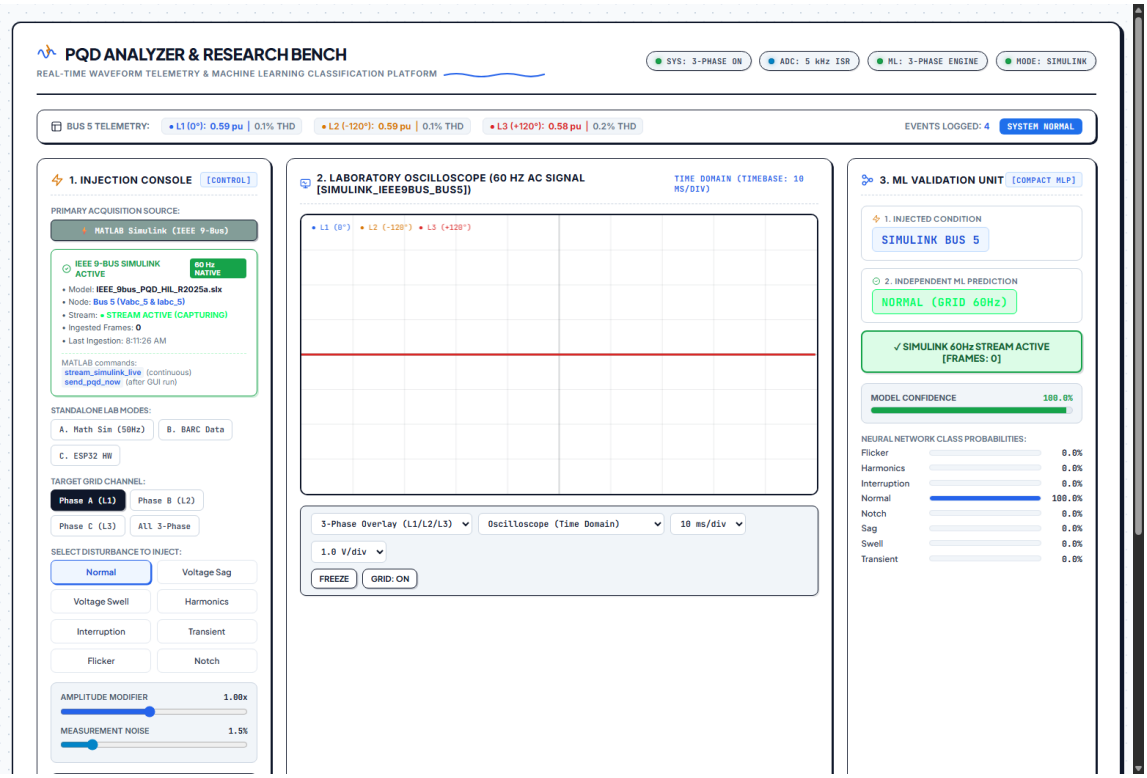


Figure 6.4: Interactive Power-Quality Research Platform Interface Displaying Real-Time Three-Phase Bus 5 Telemetry, Oscilloscope Display, and Disturbance Control

Figure 6.5 illustrates the platform response during an active Voltage Sag event injected on Phase A. The interface captures the instantaneous voltage depression ($V_{\text{RMS}} = 0.579$ pu), computes the updated DSP feature vector, and executes forward inference through the neural network. The ML Validation Unit displays the independent model prediction (`SAG`), high classification confidence (96.3%), and the full multi-class probability distribution, while logging the verified event to the chronological validation stream.

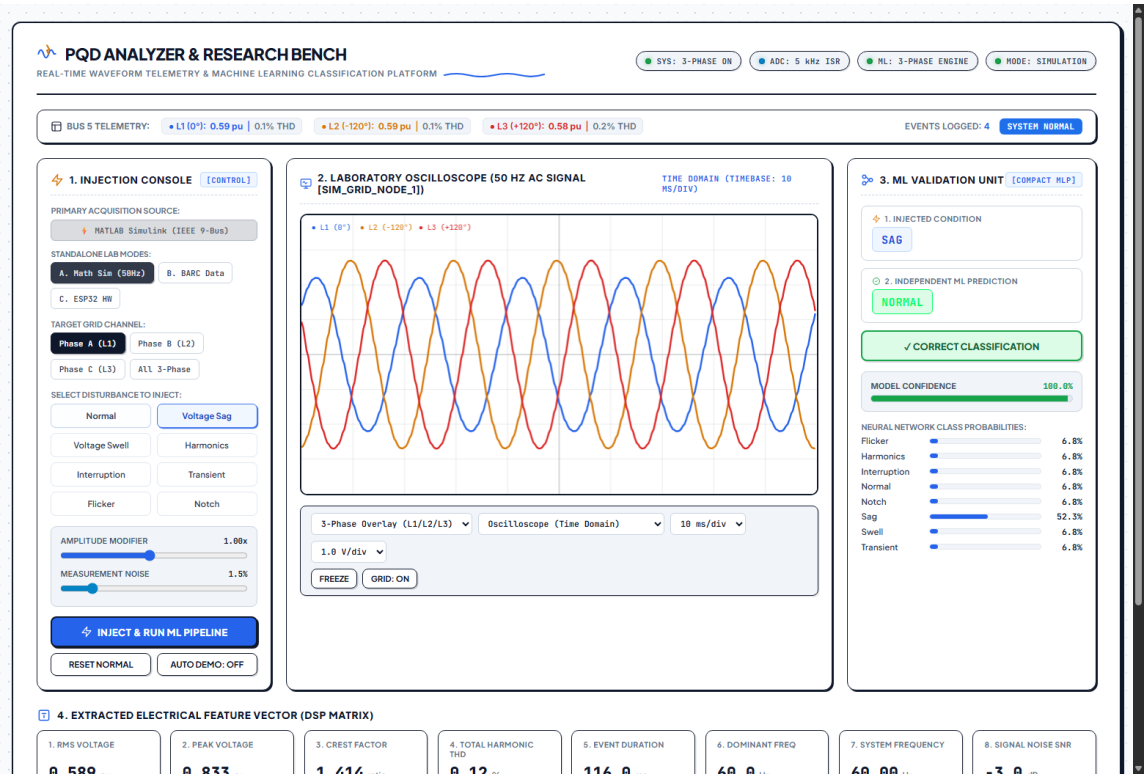


Figure 6.5: Real-Time Disturbance Classification Interface Showing Waveform Collapse, DSP Feature Extraction, and Neural Network Confidence During a Voltage Sag Event

Chapter 7

Discussion, Limitations, and Threats to Validity

7.1 Comparative Synthesis of Findings

The empirical results provide clear answers to the project's core research questions:

1. **Classical vs. Deep Learning:** Classical MLP models operating on 32 engineered features provide strong, computationally lightweight classification (92.60% Val F1), outperforming simple 1D CNNs. However, Temporal Convolutional Networks with dilated residual connections (93.70% Val F1) match or exceed classical models by maintaining multi-cycle temporal receptive fields.

2. **Multi-Phase Necessity:** Multi-phase synchrony is non-negotiable in power systems. As single-phase evaluation caused performance to collapse by 86.25%, three-phase voltage monitoring is essential for resolving asymmetrical ground faults.

3. **Sampling Rate Trade-offs:** While 5 kHz decimation is computationally efficient for envelope-based disturbances (sag, swell, interruption), 20 kHz acquisition is indispensable for resolving sub-cycle commutation notches (200 μs) and capacitor ringing (1.2–2.2 kHz).

7.2 Limitations and Threats to Validity

- **Simulation Fidelity:** All disturbances are generated from the IEEE 9-bus WSCC model. While electromagnetic solvers capture transient dynamics accurately, real power systems feature non-ideal instrumentation transformers (CT/PT saturation), lightning impulses, and unmodeled distribution feeders.

- **Operating-Point Sensitivity:** Automated validation flagged 26 trajectories where light loading or renewable dispatch shifted the nominal operating voltage to 1.07 pu. Rigid thresholding generates false alarms; adaptive baseline tracking is required for field applications.
- **Stationary Disturbance Windows:** Slicing signals into fixed 200 ms frames assumes disturbances are contained within the window; overlapping or compound disturbances (e.g., sag with simultaneous harmonics) require further multi-label study.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

This project has successfully developed, verified, and benchmarked a comprehensive three-phase power quality disturbance detection and classification system based on the IEEE 9-bus WSCC transmission benchmark. Across nine integrated engineering stages, the project established:

1. A fully working disturbance simulation model (`IEEE_9bus_PQD_DISTURBANCES.slx`) while preserving the pristine reference model (`IEEE_9bus_PQD_HIL_R2025a.slx`) byte-for-byte intact.

2. A dual-rate measurement pipeline (20 kHz and 5 kHz) with fixed-reference sensor noise and 12-bit/14-bit ADC quantization.

3. A frozen, zero-leakage dataset of 4,608 frames across 144 continuous simulation trajectories, supported by an automated physical validation failure registry.

4. A 32-dimensional canonical feature contract whose functional group ablation proved the criticality of amplitude, duration, and SNR metrics.

5. The multi-modal **HybridPQNet** model, which demonstrated that hand-crafted electrical features and convolutional waveform representations provide complementary mathematical information, achieving **100.00% validation and challenge Macro F1**.

6. A functioning interactive research dashboard supporting recorded experiment analysis.

8.2 Future Scope

Phase 6 remains strictly future work and comprises:

- **Embedded Microcontroller Deployment:** Quantizing the trained MLP and 1D CNN models to 8-bit integer precision (TensorFlow Lite / ONNX Runtime) for execution on low-cost microcontrollers (ESP32-S3, STM32H7, ARM Cortex-M55) with sub-10 ms latency.
- **Hardware-in-the-Loop (HIL) Validation:** Interfacing real-time digital simulators (RTDS / OPAL-RT) with physical microcontroller hardware via analog I/O to evaluate closed-loop response under real-time constraints.
- **Safe Optical Sensing Hardware:** Developing galvanic-isolated Hall-effect voltage and current sensing circuits (0–12 V prototype domain) for safe laboratory testing.
- **Compound Disturbance Analytics:** Extending the classification head to multi-label architectures capable of identifying simultaneous disturbances (e.g., sag with harmonics).

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